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The cortical column as a tuned receiver: a network mechanism for temporal-interference stimulation

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Abstract

Objective. Temporal-interference (TI) stimulation requires field-to-membrane access, nonlinear generation of a difference-frequency component can act on neural populations only if three stages work in series: the kilohertz field must reach a nonlinear membrane (access), the membrane must generate the difference frequency absent from the field , and amplification of that component(detection), and the circuit must amplify that weak component (amplification). We build a mesoscale model of all three stages.We connect these stages for weak perturbations of ongoing activity. **Approach.** Single-neuron TI models already demonstrate axonal ion-channel rectification . We represent candidate fast access through axonal compartments, including Following single-neuron TI models that locate rectification in nodes and the axon initial segment, we represent access by a short effective membrane time. Periaxonal circuitry motivates separate axonal and somatic voltage responses. We analyze mixing and recurrent gain We analyze detection and amplification in a Jansen–Rit column, an exact quadratic integrate-and-fire mean field mean field of quadratic integrate-and-fire neurons, and its spiking network. **Main results.** For a smooth response with a nonzero second-order term, cellular contributions combine into a beat weak fields the detector is quadratic: the beat is proportional to the product of the effective carrier amplitudeslocal carrier amplitudes, and a steady operating-point shift to their summed squares. A static rate curve mixes through its curvature; and the exact mean field recovers this description when membrane relaxation is fast give the same product law; the mean field adds frequency dependence when the membrane is slow relative to the input, leaving dynamic synapses. Otherwise mixing depends on frequency. Membrane attenuation enters once, through access. . Near a Hopf bifurcation , the network amplifies the generated beat as $1/\gamma$ within linear response, where γ is modal damping. Coupling sweeps separate this gain from leading-order beat generation. Rate modulation grows faster than the mean-rate change near resonance. Spiking simulations distinguish beat modulation below the oscillatory threshold from gamma induced when rectification crosses it. An equal-power mean-field carrier control separates induction from the beat responsethe circuit amplifies the beat as the inverse of the modal damping; with the Jansen–Rit operating point held fixed, synaptic coupling sets this gain independently of detection. The amplified response appears mainly in envelope-locked timing, with little change in mean rate. At stronger fields the steady shift alone can drive a gamma circuit across its oscillation threshold, and an unmodulated carrier of equal power does the same. **Significance.** Fast access, cellular mixing, and slow circuit resonance form a consistent candidate mechanism; its field-to-voltage transfer and response at human field strengths require calibration.

The product law makes envelope-only dosimetry conditional on transduction. Within linear response, TI and The model expresses TI efficacy as the product of three separately measurable factors. For weak fields, dose follows the product of carrier amplitudes rather than the envelope minimum, no montage can focus the steady shift inside a compartment uniform in conductivity, access, and detection, and TI shares its circuit amplifier with direct low-frequency stimulationshare the amplifier when coupling, readout, and state are matched, so resonance alone gives TI no efficiency advantage. Carrier sweeps, independent-amplitude tests, beat sweeps across states, and equal-power controls each test one stage.

Keywords: temporal interference; non-invasive neuromodulation; transcranial stimulation; neural mass model; amplitude demodulation; Hopf bifurcation; cross-frequency coupling; criticality.

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1 Introduction

Temporal-interference (TI) stimulation was proposed to reach deep brain targets while reducing stimulation of surrounding tissue [1]. It applies two sinusoids at f_1 and $f_2 = f_1 + \Delta f$ through separate electrode pairs, typically with $f_1 \sim 1$–2 kHz and $\Delta f \sim 1$–100 Hz. Where the fields overlap, their superposition is an amplitude-modulated (AM) signal with an envelope at Δf. Electrode currents can steer the region of largest envelope modulation. Whether this also localizes the neural response depends on how neurons transduce the waveform.

The difference frequency is absent from the input itself:

$$s(t) = \varepsilon_1 \cos(\omega_1 t) + \varepsilon_2 \cos(\omega_2 t), \qquad \omega_{1,2} = 2\pi f_{1,2}, \quad \Omega = \omega_2 - \omega_1 = 2\pi\,\Delta f. \tag{1}$$

Here $\varepsilon_{1,2}$ are the effective input amplitudes (related to the applied field in §2.2), Ω is the beat angular frequency, and $f_c = (f_1 + f_2)/2$ is the carrier frequency. The spectrum contains only the two applied lines (Fig. 3). A linear, time-invariant system can attenuate or phase-shift them but cannot create a new one. Passive membrane filtering therefore cannot demodulate TI [2]; a nonlinear element must mix the carriers into the low-frequency band, as in AM reception [3].

The mechanism therefore has distinct physical requirements (Fig. 1). *Access* is the conversion of the extracellular field into local membrane polarization. *Detection*, or demodulation, is the nonlinear generation of the difference-frequency component from those polarizations. *Amplification* is the response of the neuron and its recurrent circuit to the generated component. The same cellular structures can contribute to more than one stage, but strong coupling, effective mixing, and large resonant gain are different properties.

Access depends on the compartment that receives the field. A somatic charging time Somatic charging times of 10–30 ms strongly attenuates kHz polarization [4, 5] , whereas axonal geometry and myelination support different voltage responses [4, 5] imply kHz attenuation, whose magnitude also depends on cable geometry and the mode of field coupling. Esmaeilpour et al. [6] already modeled TI sensitivity in a gamma network with a 1 ms membrane time, increasing sensitivity at 0.4 ms and attributing this fast response to axons. Axonal geometry and myelination support voltage responses distinct from those at the soma. Cohen et al. [7] established a conducting periaxonal space beneath myelin that separates nodal and internodal voltage dynamics and supports saltatory conduction. This provides an anatomical basis for considering access through fast axonal compartments rather than assigning every compartment the somatic filter. Nodes and the axon initial segment (AIS) are candidate sites; the AIS also has a high density of voltage-gated sodium channels [8]. We represent this candidate route by a shorter effective membrane time. Its quantitative transfer from an applied TI field to local polarization remains to be calibrated; the measured and assumed time scales are distinguished in Table 2.

Detection requires that this polarization encounter a nonlinearity. Mirzakhalili et al. [2] demonstrated subthreshold nodal mixing in a modified McIntyre–Richardson–Grill (MRG) myelinated-axon model with Hodgkin–Huxley ion-channel dynamics. Cortical-neuron models also show axonal excitation and conduction block under stronger fields [9]. Luff et al. [10] directly observed subthreshold difference-frequency oscillations in individual neurons, including carriers near 5 kHz. Mixing persisted during synaptic blockade but was suppressed by blocking tetrodotoxin-sensitive sodium channels. These measurements establish a cellular detector without locating it within a specific compartment. In a population description, smooth cellular contributions can be combined into an effective second-order response; a static rate curve realizes that response through its curvature.

Amplification determines whether the generated beat appreciably changes activity. Intrinsic conductances such as I_h and I_M can tune a single neuron [11], as in the cell-type- and state-dependent TI responses in hippocampal neurons reported by de Palma Aristides et al. [12]. Recurrent excitatory–inhibitory circuits provide a separate source of resonance [13, 14, 15]. In slices, pyramidal and parvalbumin (PV) neurons respond differently, and most pyramidal TI responses

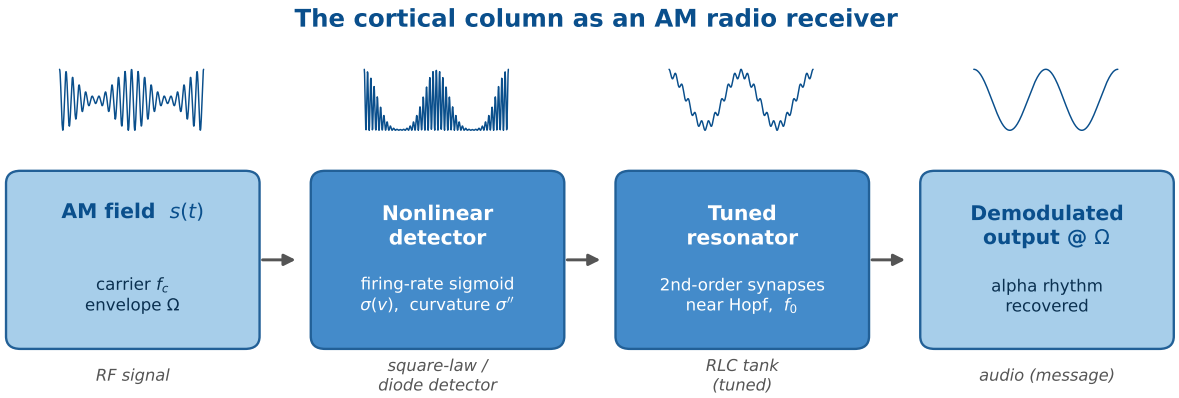


Figure 1: The cortical column as a tuned receiver. Access converts the applied field into the local polarization entering the detector. The nonlinear detector generates the beat at angular frequency Ω from a carrier near f_c; a recurrent network amplifies it near its natural frequency f_0. The diagram shows the static-sigmoid realization. In a conventional AM receiver, the tuned circuit selects the carrier before detection [3]; here the resonator selects the low-frequency output. The laminar extension also has a carrier-selective stage (§E.2).

disappear under synaptic blockade [16]. Computational studies also find network-mediated responses [17, 18, 19][17, 6]; broader accounts are reviewed in [18, 19]. Primate recordings at human-comparable fields report changes in spike timing without a corresponding mean-rate change, with weaker effects than conventional low-frequency stimulation [20]. Cellular mixing and circuit amplification can therefore contribute to the same response.

We focus on weak perturbations of ongoing neural activity. Neurons may already be spiking; weakness refers to the stimulation-induced change around that operating state. A smooth response with a nonzero second-order term gives the leading beat, and linear amplification further requires stability under the rectified mean shift. Other preparations show carrier-driven excitation or block rather than envelope extraction [21, 22, 23]. Direct carrier-evoked spiking and conduction block lie outside the weak-input calculation; finite-amplitude network examples are identified separately.

The first calculation uses a Jansen–Rit firing-rate sigmoid: its curvature mixes the carriers, and a network with second-order synapses amplifies the beat near a Hopf bifurcation. An exact quadratic integrate-and-fire (QIF) mean field connects this static description to cellular dynamics. The contribution is the separation of access, nonlinear mixing, and recurrent gain: it yields the product law for the beat and identifies the linear amplifier shared by TI and tACS. When membrane relaxation is fast relative to the input, a static population transfer curve emerges while the synapses remain dynamic; otherwise the mixing coefficient is frequency dependent. Simulations in the corresponding spiking network test the population description, and a laminar model extends the calculation to two resonant bands at low carriers.

Related neural-mass accounts use the same nonlinear mixing for endogenous cross-frequency coupling and nested amplitude modulation [24, 25]. Here the input is external, making the field-to-cell coupling an additional requirement. Our mesoscale model represents access, detection, and amplification as stages in series, each with its own test. Carrier-frequency sweeps of membrane polarization test access; independent variation of the two local carrier amplitudes tests detection; beat-frequency sweeps across operating states test amplification. Equal-power unmodulated-carrier controls separate the steady operating-point shift from the beat-locked response.

2 Theory

2.1 Population transfer and the slow-synapse limit

A neural mass model describes a population through averaged variables [26, 27]. In the Jansen–Rit (JR) family, a linear synaptic filter converts presynaptic firing rate $r(t)$ into postsynaptic potential $x(t)$:

$$\ddot{x} + 2a\dot{x} + a^2 x = Aa\, r(t),$$

with inverse synaptic time a and gain parameter A; inhibitory synapses use parameters b and B. Summed postsynaptic potentials give the mean input v, which a static transfer curve converts to firing rate [28],

$$\sigma(v) = \frac{2e_0}{1 + e^{\,\rho(v_0 - v)}}. \tag{2}$$

Here $2e_0$ is the maximum rate, v_0 the inflection voltage, and ρ the inverse voltage scale. Nonzero curvature makes this curve a mixer, while its slope sets the local gain of the recurrent circuit.

For a synapse from population Y to X, the resulting first-generation neural mass model (NMM1) equations are [26]

$$\hat{L}_{X\leftarrow Y}[u_{X\leftarrow Y}] = C_{X\leftarrow Y}\, r_Y, \qquad v_X = \Lambda[\hat{E}] + \sum_{Y:\, C_{X\leftarrow Y}\neq 0} u_{X\leftarrow Y}, \qquad r_X = \sigma(v_X).$$

$$\hat{\mathsf{L}}_{\mathsf{X}\leftarrow\mathsf{Y}}[\mathsf{u}_{\mathsf{X}\leftarrow\mathsf{Y}}] = \mathsf{C}_{\mathsf{X}\leftarrow\mathsf{Y}}\, \mathsf{r}_{\mathsf{Y}}, \qquad \mathsf{v}_{\mathsf{X}} = \Lambda[\mathsf{E}] + \sum_{\mathsf{Y}:\, \mathsf{C}_{\mathsf{X}\leftarrow\mathsf{Y}}\neq 0} \mathsf{u}_{\mathsf{X}\leftarrow\mathsf{Y}}, \qquad \mathsf{r}_{\mathsf{X}} = \sigma(\mathsf{v}_{\mathsf{X}}). \tag{3}$$

The operator $\hat{L}$ is the inverse synaptic filter, $u_{X\leftarrow Y}$ its postsynaptic potential, $C_{X\leftarrow Y}$ the connectivity, and $\Lambda[\hat{E}]$ $\Lambda[E]$ the direct polarization caused by the applied field E. This field contribution enters alongside the synaptic potentials, without first passing through a synapse.

The corresponding population nonlinearity can be derived for all-to-all coupled QIF neurons with Lorentzian-distributed excitabilities. Their exact Montbrió–Pazó–Roxin (MPR) mean field evolves the firing rate r and mean voltage v [29]; adding dynamic synapses gives NMM2 [30, 31]. In membrane-time units,

$$\dot{r} = \frac{\Delta}{\pi} + 2rv, \qquad \dot{v} = v^2 + \bar{\eta} - \pi^2 r^2 + I(t), \tag{4}$$

where $\bar{\eta}$ is the mean excitability, $\Delta > 0$ its half-width, and $I(t)$ the normalized input, including recurrent drive when present. Rate and voltage are expressed in the corresponding rescaled units. For a constant total drive $Q = \bar{\eta} + I$, setting both derivatives to zero gives $v = -\Delta/(2\pi r)$ and

$$r = F(Q) = \frac{1}{\pi}\sqrt{\frac{Q + \sqrt{Q^2 + \Delta^2}}{2}}. \tag{5}$$

Equivalently, F averages the single-neuron steady rate $f(q) = \sqrt{q}/\pi$ for $q > 0$, and $f(q) = 0$ otherwise, over the excitability distribution. Here η is the neuron's excitability and $q = \eta + I$ its constant total drive. Threshold recruitment and the concave rate increase of active cells both enter its curvature. This nonlinear curve exists without recurrent coupling. It has an inflection at $Q = \Delta/\sqrt{3}$ and an unbounded concave rise, unlike the saturating JR sigmoid.

If the population relaxes to this equilibrium faster than synaptic inputs change, NMM2 reduces to a static rate map with dynamic synapses [30, 31]. Restoring the membrane time τ_m and physical firing rate, a second-order synapse with time constant τ_s obeys

$$r(t) \simeq \frac{F(Q(t))}{\tau_m}, \qquad \tau_s^2 \ddot{s}_{\mathrm{syn}} + 2\tau_s \dot{s}_{\mathrm{syn}} + s_{\mathrm{syn}} = r(t), \tag{6}$$

where s_{syn} is the filtered rate entering the recurrent drive in Q, with the original coupling normalization retained. This is the slow-synapse limit: shortening the effective membrane time makes the

transfer appear static while the network retains its synaptic dynamics. The relevant separation is between the population relaxation time τ_{rel} and τ_s. Applying the same static map directly to a carrier of angular frequency ω_c requires the stronger condition $\omega_c \tau_{\mathrm{rel}} \ll 1$. A population can therefore be quasistatic for its synaptic inputs yet dynamic for a kHz carrier.

The transfer curve summarizes cellular nonlinearities within this regime. Its curvature combines their leading contributions to mixing; its slope converts a slow input into a rate change and enters recurrent gain. QIF supplies a derived realization through its spike-generating quadratic and excitability distribution. The generic weak-input mixing law does not require identifying each channel's contribution, although the coefficient and its frequency dependence are model-specific. Outside the static limit these contributions belong to a dynamic input–output functional. Section 2.3 calculates this response for QIF and recovers Eq. (5) in the adiabatic limit.

Cellular and circuit resonance are also distinct. Conductances such as I_h and I_M can tune single neurons [11]; the companion manuscript of de Palma Aristides et al. [12] reports cell-type- and state-dependent TI frequency selectivity in hippocampal neurons, sensitive to these conductances. Recurrent excitatory–inhibitory circuits can generate resonance without intrinsic cellular resonance [13, 14]. In the fast-membrane limit, any retained resonance belongs to the remaining synaptic circuit; modes requiring finite membrane dynamics need not persist.

The JR column couples pyramidal, excitatory-interneuron, and inhibitory-interneuron populations, with field input to the pyramidal sigmoid and local field potential (LFP) proxy $v = y_1 - y_2$(, where y_1 and y_2 are the excitatory and inhibitory postsynaptic potentials onto the pyramidal population. Its external input rate is p (s^{-1}), and C is the dimensionless connectivity scale (App. B). The laminar extension (LaNMM) [24, 32] adds coupled alpha and gamma circuits: deep pyramidal P_1, spiny-stellate (SS), and somatostatin (SST) populations, and a superficial pyramidal P_2/PV circuit. These models retain dynamic synapses and static rate curves; NMM2 and its spiking realization retain membrane dynamics as well.

2.2 From applied field to a squared-envelope beat component

We use the quasistatic electromagnetic approximation for the applied field $E(t)$ (V/m), neglecting propagation and induction. This does not make membrane charging instantaneous. Let $s(t)$ be the polarization after the cellular access response; the static population model reads

$$r(t) = \sigma(v(t) + s(t)), \qquad E(t) = E_1 \cos \omega_1 t + E_2 \cos \omega_2 t. \tag{7}$$

The field acts directly on soma, dendrites, AIS, nodes, and terminals, each with its own coupling and frequency response. In a reduced description, their contributions to the effective input are represented by parallel entry points,

$$s(t) = \sum_c \boldsymbol{\lambda}_f^c \cdot \mathbf{E}(t), \qquad \boldsymbol{\lambda}_f^c = \boldsymbol{\lambda}_0^c \, |H_c(\omega_c)|,$$

. For nearby carriers centered at $\omega_c = 2\pi f_c$,

$$\mathsf{s}(\mathsf{t}) \simeq \sum_{\mathsf{k}} \boldsymbol{\lambda}_{\mathsf{f}}^{\mathsf{k}}(\omega_{\mathsf{c}}) \cdot \mathbf{E}(\mathsf{t}), \qquad \boldsymbol{\lambda}_{\mathsf{f}}^{\mathsf{k}}(\omega) = \boldsymbol{\lambda}_0^{\mathsf{k}} \, |\mathsf{H}_{\mathsf{k}}(\omega)|, \tag{8}$$

where $\boldsymbol{\lambda}_0^c$ $\boldsymbol{\lambda}_0^k$ is the static polarization vector of compartment c k (a length), H_c and H_k its access filter , and $\omega_c = 2\pi f_c$ [26, 33]. Equation (8) uses amplitude gains and assumes aligned compartment phases; a general sum requires the complex transfers, including their relative phases. The passive compartment equation $C\dot{\mathbf{V}} + A\mathbf{V} = \mathbf{I}^0 - \mathbf{p}E$ [30] gives $\delta\mathbf{V}(\omega) = -(A + i\omega C)^{-1}\mathbf{p}E(\omega)$, with capacitance matrix C, conductance matrix A, endogenous current $\mathbf{I}^0$, and $\mathsf{C}_m\dot{\mathbf{V}} + \mathsf{G}_m\mathbf{V} = \mathbf{I}^0 - \mathbf{p}_E E$ [30] gives $\delta\mathbf{V}(\omega) = -(\mathsf{G}_m + i\omega\mathsf{C}_m)^{-1}\mathbf{p}_E E(\omega)$, where $\mathbf{V}$ contains the compartment membrane potentials, C_m and G_m are the capacitance and conductance matrices, $\mathbf{I}^0$ is the endogenous current, and $\mathbf{p}_E$ is the field-to-current

coupling. The static gain and its frequency dependence belong to this one response. A model driven by post-coupling polarization and a model driven by current through its own membrane equation must therefore count that filtering once.

For a dominant entry point, define $\lambda_f(\omega) = \lambda_0|H(\omega)|$. Then $\varepsilon_j = \lambda_f(\omega_j)E_j$ denotes the polarization reaching the nonlinearity. The direct-coupling calculation takes $\lambda_f \approx \lambda_0$ to isolate mixing and recurrent amplification; §2.6 restores frequency-dependent access. Fast axonal access motivates this limit; compartment anatomy enters the reduced model through the coupling and response time, which must be calibrated for the preparation.

Expanding σ about an operating point v^* (a Volterra/Taylor expansion, as in [25]) gives $\sigma(v^*+s) \approx \sigma(v^*) + \sigma'(v^*)\,s + \frac{1}{2}\sigma''(v^*)\,s^2 + \dots$ The truncation requires $|s| \ll 1/\rho \approx 1.8$ mV and a nonzero second derivative for quadratic mixing to be leading. Larger excursions require the full transfer curve. The linear term preserves the two input frequencies. The quadratic cross term generates their difference through the identity $2\cos\omega_1 t\cos\omega_2 t = \cos\Omega t + \cos(\omega_1+\omega_2)t$. Averaging over the carrier (the fast $\cos^2\omega_j t$ terms average to $\frac{1}{2}$, the sum-frequency term to zero),

$$\langle\sigma(v^*+s)\rangle_c = \sigma(v^*) + \tfrac{1}{4}\sigma''(v^*)\,(\varepsilon_1^2+\varepsilon_2^2) + \tfrac{1}{2}\sigma''(v^*)\,\varepsilon_1\varepsilon_2\cos\Omega t + O(\varepsilon^4), \tag{9}$$

so a line appears at exactly the envelope frequency Ω, with amplitude

$$A_\Omega \simeq \tfrac{1}{2}\,\sigma''(v^*)\,\varepsilon_1\varepsilon_2, \tag{10}$$

where $\simeq$ means to leading order in the polarizations, here and in what follows. The same expansion leaves a constant (DC) term, $\frac{1}{4}\sigma''(\varepsilon_1^2+\varepsilon_2^2)$, which any read-out of the mean rate must include.

For a dynamic cellular response with synaptic input held fixed, define the effective second-order coefficient $\chi^{(2)}$ by the complex beat amplitude $\widehat{r}_\Omega$,

$$\widehat{r}_\Omega \simeq \chi^{(2)}(\omega_2,-\omega_1)\,\varepsilon_2\varepsilon_1, \qquad r_\Omega(t) = \mathrm{Re}\left[\widehat{r}_\Omega e^{i\Omega t}\right].$$

Here $r_\Omega(t)$ is the generated rate component before recurrent feedback. With this cosine-amplitude convention, the memoryless limit is $\chi^{(2)} = \sigma''(v^*)/2$. Multiple cellular processes can contribute to the same coefficient. Small amplitude justifies the second-order expansion; it does not remove memory, so the coefficient may depend on both frequencies and the operating state. Symmetry or cancellation can make it vanish. QIF and the sigmoid realize this shared quadratic structure with different coefficients.

What the quadratic term recovers is not the envelope but its square. Writing $A(t)$ for the instantaneous envelope of the two-tone field, $A^2(t) = \varepsilon_1^2+\varepsilon_2^2+2\varepsilon_1\varepsilon_2\cos\Omega t$, the low-pass part of the squared signal is exactly

$$\mathrm{LPF}\left[s^2(t)\right] = \tfrac{1}{2}A^2(t) = \frac{\varepsilon_1^2+\varepsilon_2^2}{2} + \varepsilon_1\varepsilon_2\cos\Omega t. \tag{11}$$

This is square-law detection in the strict sense: the population transduces A^2, not A. The distinction matters because A^2 is exactly sinusoidal at Ω whereas A is not—at a matched target ($\varepsilon_1=\varepsilon_2=\varepsilon$), $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, a rectified cosine.

The amplitude-modulated surrogate. The dense perturbative sweeps below use the three-line waveform

$$s(t) = \varepsilon\,[\,1+m\cos\Omega t\,]\cos\omega_c t, \qquad \omega_c = 2\pi f_c, \tag{12}$$

with modulation depth m, because it parameterizes a sweep by one amplitude. It is not the physical signal: it carries a line at ω_c that the two-tone sum lacks. The two agree at Ω to leading order (the surrogate gives $A_\Omega = \frac{1}{2}\sigma''\varepsilon^2 m$) and are *not* equivalent beyond it. For matched genuine tones $\mathrm{LPF}[s^2] = \varepsilon^2(1+\cos\Omega t)$, whereas for the $m=1$ AM surrogate $\mathrm{LPF}[s^2] = \varepsilon^2(\frac{3}{4}+\cos\Omega t+$

$\frac{1}{4}\cos 2\Omega t$): the same Ω amplitude, but a different DC term and a spurious second harmonic at 25% of the fundamental. Since that second harmonic is the observable that distinguishes transduction classes (§2.4), every claim that turns on the waveform is run with genuine two tones : Genuine two tones are used for the demodulation demonstration (Fig. 3), the curvature and square-law controls (Fig. 6), the laminar maps (Figs. S1, S7) and the kHz demonstration. The AM surrogate is retained for the dense perturbative sweeps in (Ω, p) and (Ω, C), which read only the leading Ω component, and dense resonance and coupling sweeps; Fig. 6d verifies their open-loop agreement at Ω at one operating point. It does not establish quantitative equivalence of That agreement does not extend to the mean shift or second harmonic, the latter being a diagnostic of transduction class (§2.4). We therefore repeat the mean-shift-dependent comparisons with two tones: the timing-versus-rate figure uses them, the recurrent sweeps near the Hopf, where the TI/tACS ratios change by at most 2.2% on the tested grid, and the NMM2 endpoint remains attracting. The corresponding Results report both waveforms, whose different DC terms can shift the recurrent operating state.

In the language of [25], the quadratic cross-term $s_1 s_2$ between The quadratic cross term $\varepsilon_1\varepsilon_2 \cos\omega_1 t \cos\omega_2 t$ mixes the two fast TI tones. The same second-order response acting on a slow and a fast input (the second-order Volterra kernel) implements amplitude modulation /demodulation and the associated intermodulation linesproduces the sidebands used for endogenous amplitude modulation in [25]. Demodulation is thus itself an intermodulation product: the line at Ω is the *difference frequency* of the two applied tones (ω_1 and ω_2), generated by the same quadratic term that, acting again on an intrinsic rhythm ω_0, produces the $\omega_0 \pm \Omega$ sidebands seen at the spiking level (Fig. S12b). Equation (10) carries three predictions, all verified below: the recovered amplitude is quadratic in the drive ($A_\Omega \propto \varepsilon^2$, square-law detection); it follows the sigmoid curvature ($A_\Omega \propto \sigma''(v^*)$, hence zero at the inflection $v^* = v_0$); and, *at fixed post-coupling polarization* ε, it is independent of the carrier ω_c. This last prediction needs care: as a function of the *applied* field the response inherits the coupling $\varepsilon = \lambda_f E_0$, so the field required for a given effect rises with f_c (§2.6)—consistent with the experimental consensus, consistent with the carrier dependence of TI thresholds in [21, 22]. Carrier independence is thus a statement about the recovered beat component at fixed polarization, not about the applied field.

2.3 The exact mean field connects static and dynamic mixing

The adiabatic condition in §2.1 determines whether the static curve describes carrier mixing. Consider an uncoupled MPR population (Eq. (4), membrane time set to one), with constant drive Q and a weak added current $a\cos\omega t$. To second order in a, the carrier-averaged rate shift is $a^2 C(Q,\omega)$, where

$$C(Q,\omega) = \frac{r^*\,[\omega^2 + 3d^2 - \nu^2]}{(d^2+\nu^2)\{[d^2+\nu^2-\omega^2]^2 + 4d^2\omega^2\}}, \qquad r^* = F(Q), \quad d = \frac{\Delta}{\pi r^*}, \quad \nu = 2\pi r^*. \tag{13}$$

The fixed-point eigenvalues are $-d \pm i\nu$: d is the relaxation rate and ν the ringing angular frequency, both in membrane-time units. The coefficient follows by expanding around (r^*, v^*), with $v^* = -d/2$. The Jacobian and unit-current phasor are

$$\mathsf{J} = \begin{pmatrix} 2v^* & 2r^* \\ -2\pi^2 r^* & 2v^* \end{pmatrix}, \qquad \mathbf{h} = (\mathrm{i}\omega \mathsf{I}_2 - \mathsf{J})^{-1}\begin{pmatrix} 0 \\ 1 \end{pmatrix},$$

where I_2 is the identity matrix and the first-order response is $a\,\mathrm{Re}(\mathbf{h}e^{i\omega t})$. Writing h_r, h_v for its rate and voltage components, the carrier-averaged quadratic source per a^2 is

$$\mathbf{b}_2 = \begin{pmatrix} \mathrm{Re}(\mathsf{h}_\mathsf{r}\overline{\mathsf{h}_\mathsf{v}}) \\ (|\mathsf{h}_\mathsf{v}|^2 - \pi^2|\mathsf{h}_\mathsf{r}|^2)/2 \end{pmatrix}.$$

The mean second-order correction is $-a^2\mathsf{J}^{-1}\mathbf{b}_2$; its rate component gives Eq. (13). The limiting responses are

$$C(Q,\omega) \xrightarrow[\omega\to 0]{} \tfrac{1}{4}F''(Q), \qquad C(Q,\omega) \xrightarrow[\omega \gg d,\nu]{} \frac{F'(Q)}{2\omega^2}. \tag{14}$$

In the first limit the population tracks its static curve, whose curvature rectifies the input. In the second, the voltage excursion is approximately a/ω; its square supplies a slow drive that the population reads through F'. Above the inflection of F, these limits have opposite signs and the coefficient vanishes at $\omega_z = \sqrt{\nu^2 - 3d^2}$ (Fig. 7a). The 300 Hz and kHz carriers are on the fast side for the parameters of the pyramidal–interneuron network gamma (PING) circuit used below. That classification is parameter-dependent: a shorter effective membrane time can bring the same physical carrier into the adiabatic regime.

Membrane access is already included in this calculation. The linear voltage response is $H_v(\omega) = (i\omega + d)/[(i\omega + d)^2 + \nu^2]$. Dividing by its squared magnitude gives the mixing coefficient at fixed voltage excursion,

$$K(\omega) = \frac{C(Q,\omega)}{|H_v(\omega)|^2} = \frac{F'(Q)}{2}\,\frac{\omega^2 + 3d^2 - \nu^2}{\omega^2 + d^2}. \tag{15}$$

This coefficient tends to $F'/2$ at high frequency. The $1/\omega^2$ suppression at fixed current comes from $|H_v|^2$, the membrane filtering already represented by Eq. (8); it is not an additional detector filter. A static curve placed after the same membrane has the same leading carrier roll-off, although its mixing coefficient and sign can differ. The steady-state curvature is $F'' = 4r^*(3d^2 - \nu^2)/(d^2 + \nu^2)^3$, so the inflection occurs at $\nu^2 = 3d^2$.

For the full eight-state PING (Eq. (26)), each high-frequency current tone of amplitude a_j produces a voltage phasor $a_j/(i\omega_j\tau_E)$. The v_E^2 term therefore supplies a difference-frequency source of magnitude $a_1a_2/(\tau_E^2\omega_1\omega_2)$ inside the equation for $\tau_E\dot{v}_E$. After division by τ_E, this source is propagated through the network susceptibility at Ω. At four stable operating points, the resulting response agrees with the full second-order calculation to within 1.1% at 300 Hz, 0.1% at 1 kHz, and 0.03% at 2 kHz (Fig. 7b).

For nearby fast tones, the quadratic terms in Eq. (4) contribute at different orders: v^2 at ω^{-2}, rv at ω^{-3}, and r^2 at ω^{-4}. Table 1 compares the carrier and beat routes to the firing rate for a static curve and a dynamic population with the same membrane access.

stage	static curve, same membrane	dynamic population
point-membrane current → voltage	ω^{-1}	ω^{-1}
voltage → rate, carrier	1	ω^{-1}
current → rate, carrier	ω^{-1}	ω^{-2}
two tones → beat in the rate	$(\omega_1\omega_2)^{-1}$	$(\omega_1\omega_2)^{-1}\times$ const.

Table 1: Fast-carrier scaling with matched membrane point-membrane access, at fixed beat frequency and a nonzero leading mixing coefficient. The dynamic population integrates the carrier once more between voltage and rate; the beat is formed before that step. Prefactors depend on the operating point and beat-frequency susceptibility.

The first row is common to both detectors: the an isopotential membrane converts current to voltage with one factor of $1/\omega$. In the second row, the static curve converts voltage to rate without further frequency attenuation; its slope sets the gain. The dynamic population integrates the carrier once more in its rate equation, adding a factor of $1/\omega$. The third row combines these stages, giving a current-to-rate carrier response proportional to ω^{-1} for the static curve and ω^{-2} for the dynamic population.

The last row concerns the beat. The static curve mixes the attenuated voltage through its curvature. In the dynamic population, v^2 mixes it in the voltage equation, before the additional integration in the rate equation. Both therefore inherit $(\omega_1\omega_2)^{-1}$ from membrane access. Equal exponents do not imply equal amplitudes: the prefactors depend on the mixing coefficient and susceptibility at the fixed beat frequency Ω. For the uncoupled carrier-averaged response, Eq. (15) gives the coefficient $K(\omega)$ at fixed voltage excursion, which tends to $F'/2$ for fast carriers; it is not the full finite-Ω network prefactor. The dynamic population thus attenuates the carrier twice but

generates the beat between those stages, so its steeper carrier roll-off in the rate does not imply an additional suppression of beat generation.

2.4 Transduction class and its spatial and spectral signatures

A neural response proportional to the squared envelope is not equivalent to one proportional to the envelope. A detector implements some transduction $F(A)$ $T(A)$ of the local envelope $A(t)$, and the two laws predict different amplitude scaling, harmonic content and spatial weighting.

For a narrowband field $s(t) = A(t)\cos\omega_c t$ passing through a memoryless nonlinearity g, the low-frequency (baseband) output Y_{LF} is the average of g over one carrier cycle:

$$Y_{\mathrm{LF}}(A) = (2\pi)^{-1}\int_0^{2\pi} g(v^* + A\cos\theta)\,\mathrm{d}\theta.$$

Odd powers of A average away, leaving only even ones:

$$Y_{\mathrm{LF}}(A) = g(v^*) + \tfrac{1}{4}g''(v^*)A^2 + \tfrac{1}{64}g^{(4)}(v^*)A^4 + \cdots \tag{16}$$

A smooth memoryless response with $g''(v^*) \neq 0$ therefore transduces A^2 at leading weak-input order, whatever its detailed shape. This includes the sigmoid mechanism of §2.2. Recovering A itself requires a qualitatively different detector: strong rectification. A full-wave rectifier gives $\mathrm{LPF}[\,|A\cos\omega_c t|\,] = (2/\pi)A$ and a half-wave rectifier $(1/\pi)A$, both linear in the envelope, as does a diode-and-capacitor peak detector. A hard threshold is not one of them: a step detector with offset δ $\delta > 0$ transduces $\pi^{-1}\arccos(\delta/A)$ for $A > \delta$ and nothing below it, which is neither linear nor quadratic in A (App. D). Asymmetric channel rectification, threshold crossing and release events are candidate biological realizations of rectifiers of one kind or another. Under a linear downstream response and fixed coupling, the amplitude sweep below distinguishes a quadratic effective drive from one proportional to the envelope; the derivation of §2.2 assumes a smooth transfer, and this section asks what would show that assumption to fail.

The two classes Envelope following and square-law detection are distinguishable at a single site. At a matched target, $A(t) = 2\varepsilon|\cos(\Omega t/2)|$, whose Fourier series $|\cos u| = \frac{2}{\pi} + \frac{4}{\pi}\sum_{n\geq 1}\frac{(-1)^{n+1}}{4n^2-1}\cos 2nu$ gives a harmonic ladder in ratios $A_{2\Omega}/A_\Omega = 1/5$, $A_{3\Omega}/A_\Omega = 3/35$ and $A_{4\Omega}/A_\Omega = 1/21$. The square-law output $\frac{1}{2}A^2 = \varepsilon^2(1+\cos\Omega t)$ contains only DC and Ω. For a mixed detector $F(A) = c_1A + c_2A^2$ the mixed family $T(A) = c_1A + c_2A^2$, with $c_1 \neq 0$ and a nonzero fundamental, the pre-network ratio is

$$r_{\mathrm{det}} \equiv \left|\frac{Y_{2\Omega}}{Y_\Omega}\right| = \frac{1/5}{1 + \frac{3\pi}{4}\,c_2\varepsilon/c_1},$$

$$\mathsf{r}_{\mathrm{det}} \equiv \left|\frac{\mathsf{Y}_{2\Omega}}{\mathsf{Y}_\Omega}\right| = \frac{1/5}{\left|1 + \frac{3\pi}{4}\,\mathsf{c}_2\varepsilon/\mathsf{c}_1\right|}, \tag{17}$$

so measuring r_{det} estimates constrains $c_2\varepsilon/c_1$ directly. within this family. The inverse has two branches unless the sign of the fundamental relative to the linear term is known; for $c_2/c_1 \geq 0$ it is unique.

A smooth detector generates a second harmonic beyond quadratic order, $Y_{2\Omega}/Y_\Omega \sim (c_4/c_2)\varepsilon^2$, so the where $c_2 = g''(v^*)/4$ and $c_4 = g^{(4)}(v^*)/64$ are the coefficients in Eq. (16). The square-law signature is a *vanishing* ratio in the weak-field limit, not an exact zero. The recurrent network also multiplies each harmonic by its own gain, $r_{\mathrm{meas}} = r_{\mathrm{det}}\,|G(2\Omega)/G(\Omega)|$. At resonance($\Omega = \omega_0$) with $\mathcal{Q} = 10$, a detector ratio of 20% reads as 0.7%, making Near resonance, harmonic suppression can make an envelope follower appear square-law. The test therefore requires an *off-resonance* drive: at $\Delta f \simeq f_0/4$ both lines sit on the flat part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

An amplitude sweep tests the scaling rather than one ratio. An envelope-following detector gives $Y_\Omega \propto \varepsilon$ with the harmonic ratio pinned at 1/5; a weak square-law detector gives $Y_\Omega \propto \varepsilon^2$ with a ratio growing as ε^2. The log–log slope of the envelope-locked response against field amplitude therefore separates the classes

without requiring absolute calibration; A threshold detector is a third class: its response vanishes for $2\varepsilon \leq \delta$, and its harmonic ratio varies with amplitude after onset. A single harmonic ratio cannot distinguish all three; amplitude dependence is the ε^2 control of §4.4 is that measurement, performed in the modelprimary test.

Beyond harmonics and amplitude scaling, the transduction class fixes the spatial map. Define the excursion as $\mathcal{M}[y] \equiv \max_t y - \min_t y$ of a periodic quantity $y(t)$. The conventional TI envelope map is $\mathcal{M}[A] = 2\min(\varepsilon_1, \varepsilon_2)$, whereas the square-law detector responds to $\mathcal{M}[\frac{1}{2}A^2] = 2\varepsilon_1\varepsilon_2$, so the two are related pointwise by

$$\mathcal{M}\left[\tfrac{1}{2}A^2\right] = \max(\varepsilon_1, \varepsilon_2)\, \mathcal{M}[A]. \tag{18}$$

The square-law map is the conventional envelope-excursion map multiplied, at each point, by the dominant local polarization. It therefore does not preserve conventional TI focality. In typical montages, where Where the dominant carrier is substantially larger superficially than at depth, this reweighting degrades deep-to-superficial contrast: . As an illustrative example, at $(\varepsilon_1, \varepsilon_2) = (10, 1)$ superficially against $(0.5, 0.5)$ at a matched deep target, $\mathcal{M}[A]$ favors the superficial point by $2\times$ and $\mathcal{M}[\frac{1}{2}A^2]$ by $40\times$. The magnitude of the degradation is not fixed by this algebra and must be evaluated from the actual vector field distribution; a sufficiently weak second carrier can suppress the superficial product instead. The product law is not an account of TI focality.

The accompanying mean shift has a different spatial map. Writing $A_{\mathrm{DC}} = \sigma''(\varepsilon_1^2 + \varepsilon_2^2)/4$ for the local DC source in Eq. (9), its sign follows the curvature and is independent of carrier polarity. Carrier averaging retains the cross term as the beat; averaging over many beat cycles removes it, leaving a sum of intensities without cross-frequency cancellation. At the detector output, where $A_{\mathrm{DC}} \neq 0$, $|A_\Omega/A_{\mathrm{DC}}| = 2|\varepsilon_1\varepsilon_2|/(\varepsilon_1^2+\varepsilon_2^2) \leq 1$, with equality at matched tone magnitudes. A common access factor cancels from this local ratio. In the polarization mean-square map $\mathcal{I} = (\varepsilon_1^2+\varepsilon_2^2)/2$, the example above gives 50.5 superficially and 0.25 at depth, a 202-fold contrast; the beat/DC ratios are 0.20 and 1, respectively. The geometric restriction can be made explicit in a homogeneous, isotropic, source-free conductor. The potentials ϕ_j, each component of $\mathbf{E}_j = -\nabla\phi_j$, and the signed polarizations $\varepsilon_j = \boldsymbol{\lambda}\cdot\mathbf{E}_j$ for spatially fixed coupling $\boldsymbol{\lambda}$ are harmonic. Thus $\nabla^2\mathcal{I} = \sum_j |\nabla\varepsilon_j|^2 \geq 0$: the intensity map is subharmonic and has no strict interior maximum. A montage can shape each carrier field, but the maximum of this map lies on the region's boundary; conductivity interfaces are such boundaries. Spatially varying access, orientation, or curvature can change the physiological DC map, so the intensity result does not extend to those weights. For isotropically distributed unit orientations $\mathbf{n}$ with orientation-independent access and curvature, $\langle(\mathbf{n}\cdot\mathbf{E}_j)^2\rangle = |\mathbf{E}_j|^2/3$ and $\langle(\mathbf{n}\cdot\mathbf{E}_1)(\mathbf{n}\cdot\mathbf{E}_2)\rangle = \mathbf{E}_1\cdot\mathbf{E}_2/3$: orthogonal fields cancel the averaged beat, but retain the DC intensity. At equal local RMS polarization, an unmodulated carrier gives the same mean shift in this static quadratic limit; dynamic detectors require their frequency-dependent coefficients.

The selectivity this mechanism does supply is dynamical rather than spatial. Writing $R_\Omega(\mathbf{x})$ for the demodulated response at the envelope frequency Ω produced at location $\mathbf{x}$, and $G_{\mathbf{x}}(\Omega)$ for the local network susceptibility there,

$$R_\Omega(\mathbf{x}) \;\propto\; \sigma''(v_{\mathbf{x}}^*)\, G_{\mathbf{x}}(\Omega)\, \varepsilon_1(\mathbf{x})\, \varepsilon_2(\mathbf{x}), \tag{19}$$

a product of three factors. Field overlap sets where a difference-frequency drive is generated at all; the operating point and the network susceptibility set which of the driven populations amplifies it. Resonance does not override an unfavorable field product, but it can compensate for one. This is the radio analogy taken literally: received signal strength times receiver tuning.

Equation (19) is written for scalar projected fields. In general the coupling is vectorial and compartment-specific, $\varepsilon_i(\mathbf{x}) = \boldsymbol{\lambda}\cdot\mathbf{E}_i(\mathbf{x})$, so the mixing product $(\boldsymbol{\lambda}\cdot\mathbf{E}_1)(\boldsymbol{\lambda}\cdot\mathbf{E}_2)$ is signed: its magnitude follows the product of the projected amplitudes, while its phase inverts with either the sign of σ'' or the relative sign of the two projections. The derivations below take the collinear, scalar case.

2.5 Amplification: the synaptic low-pass and the network resonance

Synaptic filtering suppresses the carrier after the nonlinearity has generated the beat. Each JR excitatory synapse has transfer function $H_s(\omega) = Aa/(i\omega + a)^2$, a second-order low-pass with characteristic frequency $a/(2\pi) \approx 16$ Hz. It passes a 10 Hz component with relative amplitude about 0.7, while reducing a 100 Hz component to about 2.5% (Fig. 5). Provided the carrier does not itself excite a responsive mode, the loop therefore reads predominantly the generated low-frequency drive.

What turns this recovered beat component into a large, frequency-selective response is the network. Linearizing the mass about its fixed point gives a leading eigenvalue pair $\mu_\pm = -\gamma \pm i\omega_0$; near the Hopf the response is dominated by this critical mode, with damping γ and modal frequency ω_0. Below a supercritical Hopf, $\gamma \to 0^+$ and the mass is a sharply tuned (high quality-factor $\mathcal{Q} = \omega_0/2\gamma$) resonator; driving it with the demodulated beat component (10) at Ω gives a resonant response. A constant-numerator two-pole approximation gives

$$G(\Omega) \propto \left[(\omega_0^2 + \gamma^2 - \Omega^2)^2 + (2\gamma\Omega)^2\right]^{-1/2}, \qquad G(\omega_0) \propto \frac{1}{2\gamma\omega_0} \quad (\gamma \ll \omega_0), \tag{20}$$

so the response $\sim A_\Omega\, G(\Omega)$ peaks near ω_0, with an approximately Lorentzian power profile. Here ω_0 is the imaginary part of the eigenvalue; the canonical oscillator's undamped frequency is $\sqrt{\omega_0^2 + \gamma^2}$. Its peak is at $\sqrt{\omega_0^2 - \gamma^2}$ when $\omega_0 > \gamma$. The full network transfer also contains frequency-dependent input and output projections, so this expression does not fix its exact peak. The leading $1/\gamma$ growth holds near a simple Hopf with nonzero projections, within the weak-drive regime. The rectified mean shift must also leave the operating state on the stable side of the bifurcation. Above the Hopf an autonomous limit cycle appears, and the demodulated drive can entrain it within an Arnold tongue; a large lock-in amplitude alone does not establish that phase capture.

Collecting the stages, the response at the beat is

$$R_\Omega \simeq G(\Omega;\gamma)\; D\; \lambda_f(\omega_1) E_1\, \lambda_f(\omega_2) E_2, \tag{21}$$

where λ_f maps field to local carrier polarization, D converts the product of those polarizations into a beat drive, and G maps that drive to the observed response. Their normalization depends on where the generated drive enters. For JR, $D = \frac{1}{2}\sigma''(v^*)$ and G is the response to a generated rate drive. For fast-carrier QIF in the normalized units of Eq. (4), $D = 1$ for the v^2 source and G includes its conversion through the voltage and rate dynamics. Thus the population readout is included once. In these limits D is independent of carrier frequency at fixed polarization; across the dynamic crossover it acquires frequency dependence. A carrier sweep and a beat sweep therefore constrain different stages rather than a single effective gain.

The resonance does not supply the quadratic mixer. The cubic Hopf normal form responds to an in-band drive but has no second-order difference-frequency source (App. C). A large resonant susceptibility alone therefore does not establish beat generation.

2.6 Carrier access depends on the effective membrane response

Carrier independence at fixed polarization does not imply independence at fixed applied field. To quantify access , consider a first-order membrane response As an isopotential access model, consider $H_m(\omega) = 1/(1 + i\omega\tau_m)$, with effective time constant τ_m. This is the response of a point membrane to injected current; a field coupled through a cable need not have the same transfer. For nearby carriers, the two polarizations acquire nearly the same attenuation. The quadratic beat then satisfies

$$\frac{A_\Omega(f_c)}{A_\Omega^{\text{direct}}} \simeq |H_m(\omega_c)|^2 = \frac{1}{1 + (\omega_c\tau_m)^2} \sim \frac{1}{(\omega_c\tau_m)^2} \quad (\omega_c\tau_m \gg 1), \tag{22}$$

where $A_{\Omega}^{\text{direct}}$ is the beat generated with the same static coupling and no access attenuation. This filter belongs either in the field-to-polarization map or in the membrane equation that computes that polarization.

Somatic passive time constants are commonly 10–30 ms; reported human pyramidal values include approximately 12 ms [4] and 16.5 ± 3.7 ms [5]. For the illustrative $\tau_m = 16$ ms used here, the power factor $|H_m|^2$ is about 10^{-4} at 1 kHz, 2.5×10^{-5} at 2 kHz, and 10^{-6} at 10 kHz. A static nonlinearity placed after this somatic point-membrane response therefore receives strongly attenuated carriers.

Cohen et al.'s periaxonal circuit provides a reason to distinguish axonal from somatic access [7]. A conducting submyelin space, about 12 nm wide and partly sealed at the paranodes, separates the voltage across myelin from that across the axonal membrane. Their minimum fitted axonal decay was approximately 1 ms. This is a response to injected current in a distributed cable, rather than a measured transfer from a TI field. The AIS is another candidate site, with a high sodium-channel density [8]; this property supports nonlinear sensitivity but does not fix its charging time.

A shorter effective time changes carrier transmission quantitatively. The sensitivity case $\tau_m = 0.2$ ms gives $|H_m|^2 \approx 0.39$ at 1 kHz and 0.14 at 2 kHz (Fig. S5). This assumed time represents the fast-compartment hypothesis in Eq. (8), or motivates a shorter membrane time in a QIF reduction; it has not been measured under TI at the AIS or a node. In the latter description, the approach to a static transfer is governed by Eq. (6); the membrane filter must not be applied again outside the QIF equations. A short access time improves carrier transmission without, by itself, specifying the mixing coefficient or the remaining circuit resonance. Table 2 collects the response times that enter this section: measured somatic and axonal decays, the assumed fast access time, field-induced charging in a different stimulation regime, the cellular detector, and the synaptic filter acting downstream of detection. A characteristic frequency is given only for a specified filter; a measured decay is not by itself a TI cutoff.

Response	Time scale	Evidence and role
Somatic access point-membrane model	$\tau_m = 16$ ms; $f_{\text{char}} \approx 10$ Hz	Illustrative current-driven filter, informed by somatic measurements near 12 and 16.5 ms [4, 5].
Myelinated axon	minimum fitted decay ≈ 1 ms	Injected-current responses in Cohen et al.'s periaxonal circuit [7]; evidence for distinct axonal voltage dynamics.
Fast axonal access	assumed $\tau_m = 0.2$ ms; $f_{\text{char}} \approx 800$ Hz	Reduced-model sensitivity case for a fast compartment, such as an AIS or node; TI coupling and charging time require calibration.
Prior TI network model	$\tau_m = 1$ and 0.4 ms	Short effective membrane times governing kHz sensitivity in a gamma network [6]; model parameters, not measured axonal TI decays.
Field-induced charging	≈ 0.7 μs	Measured with strong microsecond field pulses in isolated neurons [23]; a different stimulation regime.
Cellular detector	channel- and state-dependent	Luff et al.'s TTX-sensitive mixing persists near 5 kHz [10]; this measures detection, not a passive time constant.
JR excitatory synapse	$\tau_s = 10$ ms; $f_{\text{char}} \approx 16$ Hz	Model filter acting on the generated rate, downstream of detection.

Table 2: Access, detection, and downstream filtering involve different response times. Measured decays, pulse-charging times, and assumed model parameters are identified separately. The characteristic frequency $f_{\text{char}} = 1/(2\pi\tau)$ is given only for the specified filters; an axonal decay is not automatically a TI cutoff. AIS, axon initial segment; TTX, tetrodotoxin.

Geometry can also alter the relevant response mode. For a passive cable, the magnitude of the

frequency-dependent space constant is

$$\ell(\omega) = \ell_0(1 + \omega^2\tau_m^2)^{-1/4}, \tag{23}$$

where ℓ_0 is its static value. At 1 kHz and $\tau_m = 16$ ms, this length is about one tenth of its static value, so compartments that are strongly coupled at low frequency can respond differently to the carrier. A somatic charging time alone therefore does not determine every extracellular-field response mode. These For a sufficiently long passive cable with a sealed end in a uniform axial field, the end polarization is proportional to this space constant. Each carrier then acquires the factor $(1+\omega^2\tau_m^2)^{-1/4}$, so the quadratic beat is attenuated by $(1+\omega_c^2\tau_m^2)^{-1/2}$, about 10^{-2} at 1 kHz and 10^{-3} at 10 kHz for 16 ms. These differ from the point-membrane estimates; soma loading and finite cable length further change the transfer. These distinct modes motivate the compartment-specific access hypothesis; their implications for TI are considered in §5.

Channel kinetics contribute to the effective dynamic mixing response. A gate filter added outside the detector must not duplicate dynamics the detector already contains, and where the detector lacks them, as QIF lacks sodium-channel kinetics, the filter needs its own model or measurement; neither a static sigmoid nor QIF specifies a universal 1 kHz gate cutoff. Equation (13) gives the QIF realization of this response. Carrier sweeps can constrain the combined access and mixing response, while beat-frequency sweeps probe the downstream amplifier.

2.7 Operating-point dependence

Because $A_\Omega \propto \sigma''(v^*)$, demodulation is governed by the curvature at the operating point, and in particular by how far v^* sits from the sigmoid inflection (Fig. 6). The Population Excitation–Inhibition Index (PEIX) of [24] is a related but distinct quantity: the sigmoid gain $\sigma'(v^*)$ normalized to its maximum $\sigma'(v_0) = e_0\rho/2$ at the inflection,

$$\mathrm{PEIX}(v^*) \equiv \frac{\sigma'(v^*)}{\sigma'(v_0)} = \mathrm{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right] \in (0, 1],$$

measures signed departure from maximal sigmoid slope:

$$\mathrm{PEIX}(v^*) = \mathrm{sign}(v^* - v_0)\left\{1 - \mathrm{sech}^2\left[\tfrac{\rho}{2}\,(v^* - v_0)\right]\right\}. \tag{24}$$

a dimensionless read-out of where v^* sits, the side given by $\mathrm{sign}(v^* - v_0)$. PEIX thus tracks σ' (maximal It is zero at v_0, positive above it, negative below it, and tends to ± 1 in the saturated tails. The normalized slope is instead $\sigma'(v^*)/\sigma'(v_0) = 1 - |\mathrm{PEIX}(v^*)|$, with $\sigma'(v_0) = e_0\rho/2$. Demodulation follows the curvature, which vanishes at the inflection v_0), whereas the demodulation gain tracks σ'' (zero at v_0, peaked and in the tails and peaks in magnitude on the flanks, and sign-reversing across it): the two are complementary, and PEIX peaks where the demodulation gainvanishes. For the logistic curve, $\sigma'' = -\rho\sigma'\,\mathrm{sign}(\mathrm{PEIX})\sqrt{|\mathrm{PEIX}|}$; the index specifies the side and saturation, but is not itself the mixing gain.

This complementarity is a property of the memoryless relation is specific to the memoryless logistic detector. In the exact mean field at a fast carrier the rectified DC term follows the slope F' and is largest at the inflection (§2.3), so the relation between PEIX and TI efficacy is model-dependent; PEIX remains a measurable read-out of the operating point. In the Jansen–Rit detector, anything shifting v^* (neuromodulation, attention, arousal, or pathology such as psychedelics or PV-interneuron dysfunction in Alzheimer's disease) scales, and can flip the sign of, the TI response; in the dynamic detector the same shifts act through F' and the susceptibility, without a sign rule. The state dependence is the falsifiable handle; the sign reversal is specific to the memoryless model.

3 Methods

3.1 Amplitude scaling and harmonic measurements

Under a linear downstream response and fixed coupling, an amplitude sweep distinguishes a quadratic effective drive from one proportional to the envelope. These tests assess the smooth-transfer assumption of §2.2 using the signatures derived in §2.4.

The response at the envelope frequency is read out with a software lock-in amplifier: multiplying $v(t)$ by sine and cosine at Ω and averaging estimates the component at that frequency. A finite window suppresses neighboring frequencies without eliminating them. With a Hann window, the in-phase (I) and quadrature (Q) components are

$$\mathsf{I} = \frac{2}{\sum_k \mathsf{w_k}} \sum_k \mathsf{w_k}\,(\mathsf{v_k} - \bar{\mathsf{v}}) \cos \Omega \mathsf{t_k}, \quad \mathsf{Q} = \frac{2}{\sum_k \mathsf{w_k}} \sum_k \mathsf{w_k}\,(\mathsf{v_k} - \bar{\mathsf{v}}) \sin \Omega \mathsf{t_k}, \quad \mathsf{A_\Omega} = \sqrt{\mathsf{I}^2 + \mathsf{Q}^2}, \tag{25}$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$; the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. Out-of-band components can leak into the estimate, particularly the DC level of the JR voltage readout ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). The finite-window projection of a constant onto the quadrature references need not vanish, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the JR kHz analysis (§2.6).

The log–log slope of the envelope-locked response against field amplitude separates the linear and quadratic laws of §2.4 without requiring absolute calibration; the ε^2 control of §4.4 is that measurement, performed in the model.

For the harmonic test, apply Eq. (25) at Ω and 2Ω over the same settled window. Their amplitude ratio estimates r_{meas}; correcting for $|G(2\Omega)/G(\Omega)|$ gives r_{det}. Inferring a mixture coefficient through Eq. (17) additionally requires the mixed family $T(A) = c_1 A + c_2 A^2$ and a known sign branch. A threshold response can give a similar ratio at one amplitude, so the amplitude sweep must also be checked. At resonance ($\Omega = \omega_0$) with $\mathcal{Q} = 10$, a detector ratio of 20% reads as 0.7%, making an envelope follower appear square-law. The test therefore requires an *off-resonance* drive: at $\Delta f \simeq f_0/4$ both lines sit on the flat part of G, where $|G(2\Omega)/G(\Omega)| \approx 1.22$–$1.25$ across $\mathcal{Q} = 3$–30, or with G measured independently by a weak sweep of direct low-frequency stimulation (transcranial alternating-current stimulation, tACS) and deconvolved.

3.2 Single-column Jansen–Rit: integration, lock-in, and controls

All symbols and units are collected in Table 1 (App. A). We integrate the single-column JR model of §2.1 (Eqs. (1), parameters in Table 2) as six first-order ODEs. The TI field enters only the pyramidal output sigmoid, Eq. (1), as $s(t)$; the read-out $v = y_1 - y_2$ is *not* fed the field directly, so any power at Ω in v is produced by the nonlinearity and the recurrent loop, not by trivial pass-through. Unless stated otherwise $f_c = 100$ Hz and $m = 1$.

We integrate with a fixed-step fourth-order Runge–Kutta (RK4) scheme, $\Delta t = 2 \times 10^{-4}$ s (50 samples per carrier cycle at 100 Hz; reduced to 10^{-4} s for carrier sweeps up to 300 Hz). The kHz-carrier runs of Fig. S9 use $\Delta t = 2 \times 10^{-5}$ s for the carrier sweep to 4 kHz (25 samples per carrier cycle at 2 kHz, 12.5 at 4 kHz) and 1.5×10^{-5} s for the 2 kHz output spectrum. Halving each step changes the reported lock-in amplitudes by less than 1%. The 10 kHz figures quoted in §2.6 are analytic evaluations of Eq. (22), not simulations.

The resonance sweep in Fig. 4 uses 10 s of settling and a 20 s measurement window. Long windows reduce transients from weakly damped modes and average beating when autonomous activity persists. Other runs use the windows specified in their scripts. The two-tone 2 kHz example (Fig. S6) uses $p = 400$ s^{-1}, $C = 135$, and 0.5 mV per tone with $\tau = 0.2$ ms. From a zero initial state, it settles for 4 s and records 4 s at $\Delta t = 1.5 \times 10^{-5}$ s, sampling before each RK4 step. Its amplitude spectra use plain-mean subtraction and a Hann window. Parameter sweeps are

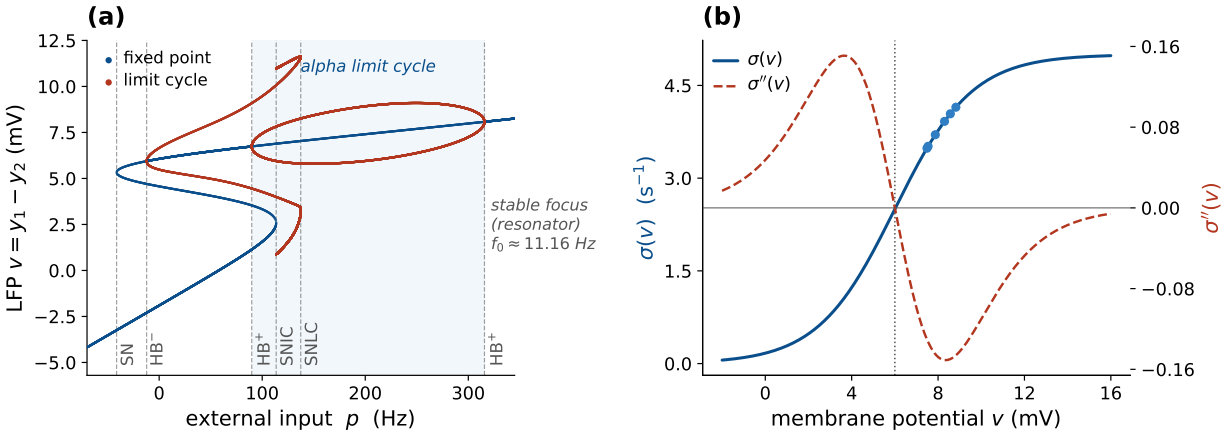


Figure 2: The Jansen–Rit operating regime: bifurcation and sigmoid curvature. *(a)* JR bifurcation diagram from numerical continuation—the LFP $v = y_1 - y_2$ along the fixed-point branch (blue) and the min/max of the limit-cycle branches (red) vs. external input p, with *solid* stable and *faded* unstable. The alpha limit cycle (shaded) is bounded by a saddle-node-on-invariant-circle (lies between the Hopf points at $p \approx 90$ and 316 s^{-1}. The SNIC , at $p \approx 114$) and terminates the supercritical large-amplitude branch. At the upper Hopf($p \approx 315$, $f_0 \approx 11.1$ $f_0 \approx 11.16$ Hz); beyond it the Hopf the fixed point is a stable focus, the near-Hopf resonator used here. Further codim-1 points are marked (saddle-node SN, Hopf HB$^{\pm}$, saddle-node of limit cycles SNLC). *(b)* The sigmoid $\sigma(v)$ and its curvature $\sigma''(v)$, which sets the demodulation gain and vanishes at the inflection v_0. Dots mark the operating points reached by the closed-loop model for the p values used here.

vectorized: the state is an $(N, 6)$ array advanced simultaneously for all N parameter combinations, so a full resonance map runs in seconds.

The full bifurcation diagram (Fig. 2) is computed by numerical continuation of the fixed-point and limit-cycle branches. We independently locate the operating supercritical Hopf by solving, for each p, the self-consistent scalar equation for the resting fixed point $v = y_1 - y_2$, forming the 6×6 Jacobian numerically and diagonalizing it: the Hopf is the p at which the leading eigenvalue pair crosses the imaginary axis with $\mathrm{Im}\,\mu_+$ in the alpha band. This gives $p \approx 315$ Hz and $f_0 = \mathrm{Im}\,\mu_+/2\pi \approx 11.1$ Hz—consistent $p_H \approx 315.70$ s^{-1} and $f_0 = \mathrm{Im}\,\mu_+/2\pi \approx 11.16$ Hz, consistent with the continuation and with the JR bifurcation the JR analysis of [34]. For $p > 315$ The 11.1 Hz demonstration beat is near this resonance. For $p > p_H$ the fixed point is a stable focus(resonator); between . The alpha cycle lies between Hopf points near $p = 90$ and 316 s^{-1}; it coexists with a lower stable fixed point up to the saddle-node on an invariant circle (SNIC, $p \approx 114$)and the Hopf the column free-runs on the alpha limit $p \approx 114$), then with a large-amplitude cycle up to its saddle-node near $p = 137$. The SNIC terminates that large-amplitude branch, not the alpha cycle.

The response at the envelope frequency is read out with a software lock-in amplifier: multiplying $v(t)$ by sine and cosine at Ω and averaging estimates the component at that frequency. A finite window suppresses neighboring frequencies without eliminating them. With a Hann window, the in-phase (I) and quadrature (Q) components are

$$I = \frac{2}{\sum_k w_k} \sum_k w_k\,(v_k - \bar{v}) \cos \Omega t_k, \quad Q = \frac{2}{\sum_k w_k} \sum_k w_k\,(v_k - \bar{v}) \sin \Omega t_k, \quad A_\Omega = \sqrt{I^2 + Q^2},$$

with w a Hann window and $\bar{v} = \sum_k w_k v_k / \sum_k w_k$ the windowed mean. For a pure tone $v = A\cos(\Omega t + \phi)$ the product-to-sum identity gives $I \to A\cos\phi$ and $Q \to -A\sin\phi$, so $A_\Omega \to A$; the factor $2/\sum_k w_k$ is the normalization that returns the physical amplitude, and A_Ω equals twice the magnitude of the windowed Fourier coefficient at Ω. Out-of-band components can leak into the estimate, particularly the DC level of v ($\bar{v} \sim 8$ mV, far larger than the demodulated signal). The finite-window projection of a constant onto the quadrature references need not vanish, so we subtract $\bar{v}$ explicitly; this makes the metric accurate down to the 10^{-8} level needed for the kHz analysis (§2.6). For the For the open-loop rate readout, the signed in-phase component I recovers $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ including its sign, which we use to verify the curvature law (Fig. 6).

We compare the *open-loop detector*, the firing-rate sigmoid at a fixed operating point v^* without recurrence, with the *closed-loop* column, the full recurrent Jansen–Rit with its feedback. The open-loop detector passes the two-tone field through σ at v^* and applies Eq. (25), isolating the pure demodulation law (Figs. 5, 6, S5). The *bifurcation diagram* records the field-free min/max of v vs. p, each run initialized at its fixed point plus a small kick. The *two-dimensional map* evaluates A_Ω over an (Ω, p) grid. We verify the mechanism three ways: the square law (A_Ω vs. ε on a well-damped focus, expecting a log–log slope of 2); a linearization control (replacing σ in Eq. (1) by its tangent at v^*, which must abolish the Ω response); and a check of σ', σ'' against finite differences. Code and reproduction limits are listed under *Code and data availability.*

3.3 Laminar two-band model (LaNMM)

We complement the single-column analysis with simulations of the laminar neural mass model (LaNMM) of §2.1. We integrate the full 28-variable column [24, 32], with the field entering as in the single JR column, as a membrane polarization added inside the sigmoid argument of the target population,

$$\varphi_X(t) = \sigma\big(v_X(t) + s(t)\big), \qquad X \in \{P_1, P_2\},$$

with $s(t)$ the AM or two-tone field of §2.2 (amplitude ε in mV, beat $\Omega = 2\pi\Delta f$, carrier $\omega_c = 2\pi f_c$) and all three external drives held flat at their baselines (Fig. S2). A perturbation of an external input *rate* would require clipping at zero once its excursion exceeds the baseline, and a hard clip is a half-wave rectifier—a diode detector placed upstream of the sigmoid, which would inject a line at Δf into the drive before the population nonlinearity received it. A signed membrane polarization carries no such constraint and none is applied. Because the LaNMM and JR sigmoids share the voltage scale $1/\rho \approx 1.8$ mV, ε is directly comparable between the two models.

Integration is a vectorized fixed-step RK4 ($\Delta t = 10^{-4}$ s) advancing the whole parameter grid simultaneously, as in the JR sweeps. The maps report the alpha-band (8–12 Hz) power of the P_1 and P_2 membrane potentials as the mean squared magnitude of the Hann-windowed fast Fourier transform (FFT) over the band, in arbitrary units, over a post-transient window. This measure includes autonomous and driven activity and does not isolate a phase-coherent response to stimulation.

Two sweeps probe the column: an amplitude–beat sweep over $(\varepsilon, \Delta f)$ at fixed carrier, and a carrier–beat sweep over $(f_c, \Delta f)$ at fixed ε, built on the LaNMM module of [24]. The maps (Figs. S1, S7) show the *signed* stimulation-induced change, alpha power minus the unstimulated ($\varepsilon = 0$) baseline, on a zero-centered scale; over most of the P_1-driven grids the change is negative. For the alpha resonance we instead drive the deep P_1 loop directly and sweep its input against Δf, reading the lock-in (Eq. (25)) of v_{P_1} at Δf.

The additional carrier comparison in Appendix E uses the lock-in of v_{P_1}, averaged at each carrier over 17 beats: 7–9 and 12.25–14 Hz in 0.25 Hz steps. These lie more than 1.5 Hz from the field-free alpha peak near 10.6 Hz. We denote this arithmetic mean by $S(f_c)$. The carrier grid is 25–80 Hz in 1 Hz steps, with equal tone amplitudes $\varepsilon = 1.5$ mV for P_2 drive and 0.5 mV for P_1 drive. After 2 s of settling, we compare 3 and 10 s measurement windows. Field-free controls give mean background amplitudes of 0.0048 and 0.0012 mV over the same beats. This background includes a persistent component near 8 Hz, so it is not solely finite-window leakage. The statistic compares carrier profiles within each drive condition; the unequal amplitudes do not support a direct sensitivity comparison between targets.

3.3 Exact next-generation mean field (NMM2 PING)

The NMM2 PING circuit couples excitatory and inhibitory MPR populations through second-order synapses [29, 30, 31]. We use the Rosetta Stone formulation [26], which places this derived mean field on the same footing as JR and LaNMM (§2.1). For population $\alpha \in \{E, I\}$, r_α is the firing rate, v_α the mean potential, and τ_α the membrane time:

$$\begin{aligned} \tau_E \dot{r}_E &= \Delta/(\pi\tau_E) + 2 r_E v_E, & \tau_E \dot{v}_E &= v_E^2 + \bar{\eta} - (\pi\tau_E r_E)^2 + \tau_E C\,(s_E - s_I) + I_E(t), \\ \tau_I \dot{r}_I &= \Delta/(\pi\tau_I) + 2 r_I v_I, & \tau_I \dot{v}_I &= v_I^2 + \bar{\eta} - (\pi\tau_I r_I)^2 + \tau_I C\,(s_E - 2 s_I), \end{aligned} \tag{26}$$

with second-order synaptic activations $s_E = \hat{K}_a[r_E]$ (excitatory, time constant τ_a) and $s_I = \hat{K}_g[r_I]$ (inhibitory, τ_g), each obeying the same critically-damped operator as the JR synapse, $\tau^2 \ddot{s} + 2\tau \dot{s} + s =$

r; here C is the global synaptic gain and the weights $(A_{ee}, A_{ei}, A_{ie}, A_{ii}) = (1, 1, 1, 2)$ set the E and I rows: the E membrane sees $C\,(s_E - s_I)$ and the I membrane $C\,(s_E - 2s_I)$, the only asymmetry being the doubled $I \to I$ weight $A_{ii} = 2$.

The common background drive $\bar{\eta}$ is the bifurcation parameter: raising it past a supercritical Hopf at $\bar{\eta}_H = 1.110$ opens a gamma limit cycle (the gamma counterpart of the JR alpha Hopf). The applied field enters the excitatory voltage equation as a current-like forcing, $I_E(t) = P\,E(t)$ in the convention of the exact mean field [30], with the dipole coupling P instantaneous, as two genuine tones at $f_c \mp \Delta f/2$ (carrier $f_c = 300$ Hz, envelope Δf swept through the gamma band) of amplitude A_f in the dimensionless MPR units of v_E. The E membrane $\tau_E \dot{v}_E$ integrates this forcing into the carrier-frequency excursion $A_f/(\omega\tau_E)$ that the exact quadratic v_E^2 term rectifies, with no postulated sigmoid; the PING as run therefore computes membrane access with $\tau_m = \tau_E$.

We integrate with fixed-step RK4 ($\Delta t = 0.05$ ms) and read out the lock-in (Eq. (25)) of r_E at Δf over the $(\Delta f, \bar{\eta})$ plane. Parameters (Rosetta Stone PING [26]): $\Delta = 1$, global coupling $C = 15$, $\tau_E = 15$, $\tau_I = 7.5$, $\tau_a = 10$, $\tau_g = 2.5$ ms, with $\bar{\eta}$ swept through the supercritical Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz), located by bisection on the leading eigenvalue of the mean-field Jacobian. The simulations retain these finite membrane times; Eq. (6) characterizes the fast-membrane limit.

As in JR, either background drive or coupling can bring the circuit to its Hopf. The resonance map sweeps $\bar{\eta}$ (Fig. S8); the coupling sweep, called the J-curve, varies $J \equiv C$ at fixed $\bar{\eta}$ toward J^*, located by bisection on the leading Jacobian eigenvalue, and reads the lock-in peak of r_E over Δf (Fig. S10). Unlike the JR J-curve, this sweep does not pin the operating point. In the fast-carrier approximation the leading voltage source is fixed by the carrier excursion (§2.3). The four-state check bounds the error of that approximation at the tested points; it does not establish source invariance over the full J sweep. The plotted normalization therefore uses the leading fast-carrier source, while the finite-carrier response can retain At $\bar{\eta} = 0$, we track the eigenpair that reaches the Hopf by continuation and define γ from that critical pair, even where another mode dominates the response. The linear reference includes the operating-state dependence. dependence of the carrier response and the recurrent transfer. We compute its second-order source from the PING quadratic terms at the central carrier, 300 Hz, and apply the full linearized network response over 15–90 Hz in 0.025 Hz steps. Using the distinct AM sidebands changes the peak gain by less than 0.1% across this sweep.

The simulated coupling sweep uses the AM surrogate, $A_f = 8$, with 5 s settling and a 4 s window. We search around each linear resonance, extending the frequency grid until each sampled maximum is bracketed, and compare amplitudes $A_f = 2, 4, 8$ at nine couplings. Local parabolic interpolation changes the sampled peaks by less than 1.3%. The figure scales the linear reference to the median measured gain over $14.5 < J < 16$; the amplitude check instead normalizes peak response divided by A_f^2 and linear gain to its value at $A_f = 2$, $J = 15.53$. These normalizations compare gain variation, rather than establishing invariance of the detector coefficient.

We check the stability of the driven J-curve endpoint by integrating the full variational matrix along its trajectory. At carrier 300 Hz and beat 50.5 Hz, the AM input repeats every 2 s. After 8 s of settling, we compute the perturbation multipliers over that period and repeat over the next period. Moduli below one indicate decay of perturbations to the periodic response. Two-tone controls at $J = 16.5663$ and 16.5901, both with a 50.5 Hz beat, use 5 s settling, direct perturbation decay, and an 8 s finite-time Lyapunov estimate.

3.4 QIF spiking network (microscale of NMM2)

NMM2 is the exact infinite-population mean field of all-to-all QIF neurons [29, 26, 31]. We integrate that network directly to compare beat-locked and ongoing activity at the spike level (§4.8): $N = 8000$ QIF neurons per population, parameters identical to the PING mean field (Eq. (26)), with finite spike threshold $V_{\text{peak}} = 100$ and reset $-V_{\text{peak}}$ in model voltage units, excitabilities $\eta_j \sim$ Lorentzian$(\bar{\eta}, \Delta)$ sampled by deterministic quantiles, second-order synapses driven by the empirical

population rates, and a genuine two-tone field entering the excitatory membranes. Spiking and threshold dynamics are where the AM surrogate and the physical waveform cannot be assumed equivalent beyond perturbation theory, so this run uses two tones throughout.

The common background drive $\bar{\eta}$ places the field-free mean field below or above a gamma Hopf at $\bar{\eta}_H = 1.110$ ($f_0 = 54.0$ Hz). We use $\bar{\eta} = 0.85$ with a 55 Hz beat and $\bar{\eta} = 1.5$ with a 42 Hz beat. Both runs have carrier $f_c = 300$ Hz and amplitude $A_f = 35$ per tone. These are finite-amplitude illustrations: the fast-carrier estimate gives a voltage excursion $A_f/(\tau_E\omega_c) \approx 1.2$ per tone in the model's voltage units. The field can therefore alter the dynamical regime, even when the unstimulated state is below HopfIts rectified excitatory-drive shift is about 1.5, compared with 0.12 at $A_f = 10$. At $\bar{\eta} = 0.85$, the Hopf threshold for a drive shift applied only to the excitatory population is $\delta\eta_E^* \approx 0.286$. The larger field is therefore expected to cross this threshold, whereas the smaller one remains below it. We label the conditions by their field-free state and separate the beat-locked component from other gamma activity. The spectra and rate folds do not by themselves establish phase capture (§F).

We also run both conditions at $A_f = 10$ per tone with the same measurement settings. For $\bar{\eta} = 0.85$, separate checks double the population size, double the measurement window, or halve the integration step; further runs raise V_{peak} to 200 and 400 at both time steps. An equal-RMS mean-field control replaces the two tones by one 300 Hz carrier with peak amplitude $\sqrt{2}A_f$, using the same duration: both inputs have mean-square amplitude A_f^2 over these windows. The carrier comparison starts from the field-free equilibrium and uses 800 ms settling and 800 ms measurement, with a 3200 ms window to check spectral leakage at the larger amplitude.

For two nearby carriers, the fast-carrier approximation gives a constant shift of the excitatory drive, $\delta\eta_E \simeq A_f^2/(\omega_c\tau_E)^2$, from the quadratic voltage term. We refine this estimate using the full PING linear carrier responses and their second-order DC source, then solve for the carrier-averaged equilibrium and its stability, including the source's dependence on that equilibrium. This approximation isolates the mean shift; direct two-tone integrations retain the beat forcing and provide a separate check.

The network is integrated with Euler steps of 0.0025 ms, settled for 800 ms (over four field-free relaxation times at $\bar{\eta} = 0.85$, where the mean-field relaxation time is 176 ms) and measured over an integer number of beat cycles, 44 at 55 Hz and 34 at 42 Hz (800 and 810 ms), removing the phase-coverage bias from incomplete beat cycles. Envelope-locked timing is quantified by the quadrature lock-in A_Ω of $r_E(t)$ at Δf, reported as the modulation depth $A_\Omega/\langle r_E\rangle$ (a stable measure; the off→on lock-in ratio would divide by the field-free floor), and per neuron by the pairwise phase consistency of each neuron's spike phases at Δf [35], which is unbiased in the spike count, over neurons with at least ten spikes in the window.

In the $A_f = 35$ convergence checks, halving the time step changes the lock-in and the pooled vector strength by under 1%; doubling the spike-detection threshold changes the lock-in by about 1% and the mean rate by 2%, but the pooled vector strength at $\bar{\eta} = 0.85$ by about 45%, so phase concentration, unlike rate modulation, is sensitive to how spike times are defined.

3.5 Coupling sweeps, timing, and the tACS control (Jansen–Rit)

The coupling sweeps of Figs. 9 and S11, and the transcranial alternating-current stimulation (tACS) control of Fig. S3, use the single JR column with the field-free operating point v^* pinned: for each global connectivity C we solve the field-free fixed-point relation analytically for the external drive $p(C)$ that holds v^* (hence $\sigma''(v^*)$) fixed, vary $C\,(= J)$ toward the alpha Hopf, and read γ, ω_0 from the 6×6 Jacobian. The field is the AM surrogate of Eq. (12) for TI (Fig. 9; AC lock-in vs DC mean rate for , genuine two tones for the timing comparison (Fig. S11), or a direct in-band sinusoid $s(t) = \varepsilon\cos\Omega t$ ($\varepsilon = 0.1$ mV) for the tACS control (Fig. S3), each read out by the lock-in of v at Δf. , except in the timing comparison, which reads the modulation and mean of the same pyramidal rate $r_P = \sigma(v + s)$. That comparison uses 18 s settling and a 20 s window for each waveform. The Figure 9–Figure S3

response-ratio comparison uses $v^* = 8.30$ mV, TI surrogate amplitude 0.3 mV with $m = 1$ and $f_c = 100$ Hz, and tACS amplitude 0.1 mV, driven at each field-free modal frequency. The original TI amplitudes are interpolated logarithmically in C onto the tACS grid. For the waveform comparison, both the surrogate and two-tone TI are rerun on that grid with 12 s settling and a 6 s window. Pinning fixes the background curvature; it does not cancel the stimulation-induced mean shift. Additional controls at $C = 110$, 133.9091, and 136.3 use a tACS amplitude of 0.0143039 mV, matched to the leading TI beat source. They record mean voltage, mean rate, and lock-in with 12 s settling and a 6 s window, repeated with 60 s settling and a 20 s window; the latter settings also test genuine two tones.

3.6 Vocabulary: forced response, entrainment, and gating

Forced response denotes the component driven at the beat Δf, quantified by the lock-in amplitude (Eq. (25)). A purely forced regime requires a stable operating state under stimulation; being below Hopf with the field off is insufficient. *Entrainment* is reserved for its dynamical-systems meaning, frequency capture of a *self-sustained* oscillator [36]; in the maps we screen for it with the dominance criterion $|f_{\text{out}} - \Delta f| < 0.3$ Hz on the dominant output frequency. This identifies frequency dominance, not phase capture. *Gating* is amplitude modulation of a persisting rhythm without frequency capture [37, 38]. Capture and gating are dynamically different, and the same word would otherwise cover both. The QIF rate fold measures beat-locked firing, which does not by itself establish gating of gamma amplitude. Appendix F gives the metrics and the frequency-dominance protocol.

4 Results

The Jansen–Rit column and the exact next-generation mean field (NMM2) both generate a difference-frequency component and amplify it near a Hopf bifurcation. The former uses a memoryless sigmoid; the latter derives the nonlinearity from QIF dynamics. Their detector coefficients differ, but both separate beat generation from recurrent gain. Coupling controls that gain at fixed generated drive in the JR column (§4.6), and resonance amplifies beat-locked modulation more strongly than the mean rate (§4.7). Finite-amplitude QIF simulations then show how a beat-locked component coexists with gamma activity at the spike level. The laminar two-band column extends the analysis to carriers of 25–80 Hz and drives up to 3 mV (App. E).

4.1 The Jansen–Rit sigmoid demodulates: an alpha line from a carrier-only input

An input containing only the two tones $f_c \mp \Delta f/2$ produces an output oscillation at their 11.1 Hz beat, near the column's alpha resonance (Fig. 2). The alpha peak is absent from the input spectrum (Fig. 3).

4.2 Near the Hopf, the demodulated beat drives a sharp resonance

In the stable-focus regime ($p > 315 p > p_H$), the response is approximately Lorentzian and its peak grows from 0.23 to 0.43 to 0.70 mV toward the Hopf (Fig. 4), as expected from the $1/\gamma$ susceptibility. Without a drive the focus is silent at Ω; the nonlinearity generates that component and recurrence amplifies it. On the oscillatory side ($p < 315$)alpha branch below p_H, the column free-runs at f_0, so the lock-in can contain both driven and autonomous activity. The dominant output frequency f_{out} then identifies a frequency-dominance region (Fig. S4), without establishing phase capture.

4.3 Carrier independence and the kHz roll-off

At fixed post-coupling polarization the demodulated response is flat across the carrier band, for both the open-loop detector and the closed-loop column near the Hopf (Fig. 5). As a function of

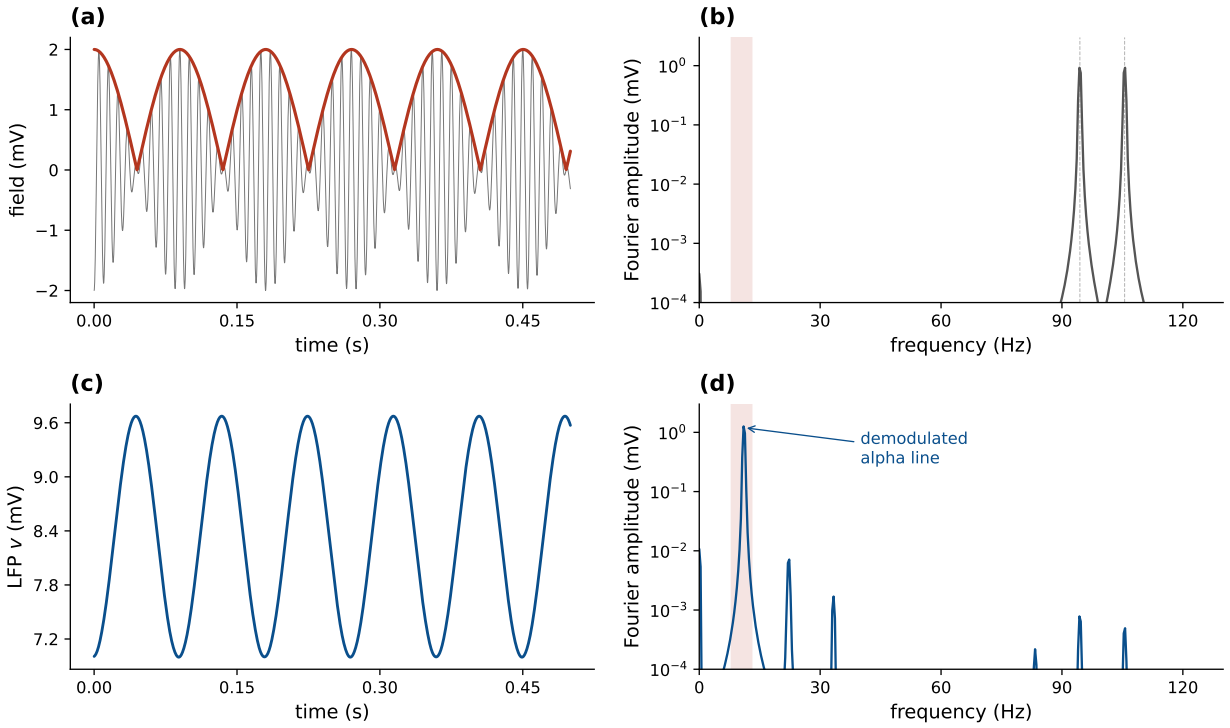


Figure 3: An alpha rhythm is synthesized from a two-tone TI input (Jansen–Rit below Hopf). The field is the physical TI waveform, two equal tones at $f_{1,2} = f_c \mp \Delta f/2$, not the amplitude-modulated surrogate. *(a)* The field $s(t)$ and its envelope $A(t) = 2\varepsilon|\cos(\Omega t/2)|$—a rectified cosine, not a sinusoid; it is A^2 that is sinusoidal at Ω (§2.2). *(b)* The input amplitude spectrum has lines only at f_1 and f_2 (94.45 and 105.55 Hz, $\Delta f = 11.1$ Hz) and no third line at the carrier; in the alpha band (shaded) it is 1.6×10^{-8} of the tone amplitude, finite-window leakage rather than physical power. *(c)* The output $v = y_1 - y_2$ oscillates at the envelope frequency. *(d)* The output spectrum has a strong peak at the beat, 11.1 Hz, synthesized by the sigmoid and absent from the input.

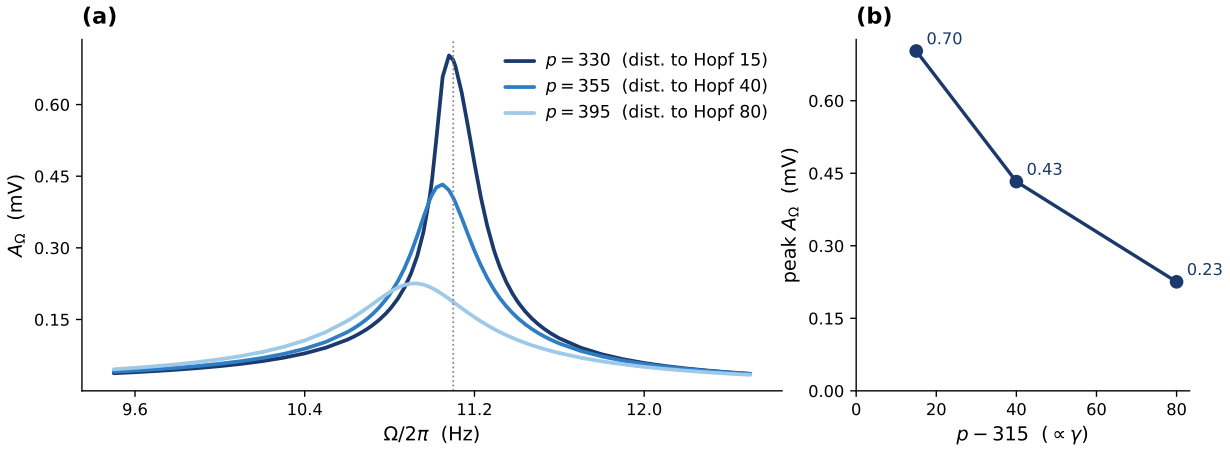


Figure 4: The demodulated beat drives a resonance that sharpens toward the Hopf (Jansen–Rit). Fixed-point regime ($p > 315 p > p_H$, stable focus): lock-in response at Ω vs. envelope frequency ($\varepsilon = 0.3$ mV). *(a)* The demodulated drive forces an approximately Lorentzian resonance near $f_0 \approx 11.1 f_0 \approx 11.2$ Hz. The measured finite-amplitude peak shifts from 11.08 to 10.93 Hz across the curves as the operating state changes; Eq. (20) describes its near-Hopf scaling, not its exact location. The response carries no power at Ω in the absence of drive, so this is forced gain. *(b)* The peak height grows as the Hopf is approached ($p \to 315$, distance to Hopf $\to 0 p \to p_H$ from above). Three points cannot establish a power law; the $1/\gamma$ scaling is quantified over a decade, together with its saturation, in Fig. 9(c). The above-Hopf regime, with autonomous alpha activity, is characterized by output frequency in Fig. S4.

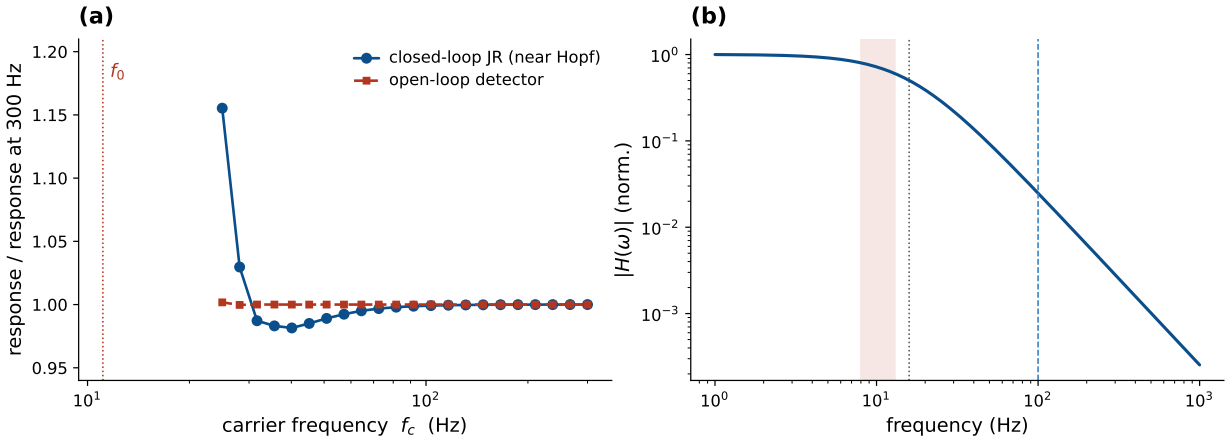


Figure 5: The demodulated response is carrier-independent at fixed polarization (Jansen–Rit). *(a)* The demodulated response is nearly flat across carrier frequency $f_c = 25$–300 Hz for the open-loop detector and for the closed-loop JR near the Hopf, with deviations as f_c approaches the natural band. Each curve is normalized by its own response at 300 Hz: 0.00670 s^{-1} open loop and 0.512 mV closed loop. Their ratio is a rate-to-voltage transfer of 76.4 mV s. The open-loop carrier independence follows to leading order from Eq. (10). *(b)* The second-order-synapse low-pass $|H_s(\omega)|/|H_s(0)|$ passes the envelope ($\sim$0.7) and suppresses the 100 Hz carrier ($\sim$2.5%). Dotted line: synaptic corner $a/2\pi \approx 16$ Hz; shaded band: envelope (8–13 Hz); dashed line: 100 Hz carrier.

the *applied* field, by contrast, the membrane low-pass makes the response roll off as $1/f_c^2$ above the corner in the point-membrane model (Fig. S5): passive somatic demodulation is suppressed four to six orders of magnitude at kHz in the 16 ms access model, whereas the . Its 16 ms response gives beat attenuation of about 10^{-4}–10^{-6} at 1–10 kHz; the cable-end case of Eq. (23) instead gives about 10^{-2}–10^{-3}. The assumed 0.2 ms point-membrane response retains appreciable carrier access.

We also drive the full recurrent column with a kHz field, passing it through an explicit membrane low-pass (an extra first-order state, time constant τ) for each coupling route before it modeled compartment before its polarization reaches the sigmoid. The column demodulates and resonates at the recovered alpha in the fast-access case ($\tau = 0.2$ ms), as Eq. (22) predicts (Fig. S9a). Driving the fast-element route with two *genuine* tones at $f_{1,2} = 1994.54$ and 2005.46compartment with $\tau = 0.2$ ms using two genuine tones at $f_{1,2} = 1994.543$ and 2005.457 Hz, centered near at 2 kHz and beating at 10.91$10.914$ Hz, the finite-window input spectrum has a maximum amplitude of 2.2×10^{-13} mV in the 8–14 Hz band, while the output carries a line at the beat (Fig. S6), with the attenuation set by the analytic $|H_m|^2 = 0.137$ at 2 kHz for $\tau = 0.2$ ms. The two-tone result therefore quantifies the fast-access hypothesis under the specified response time and coupling. Limited sensitivity to unmodulated kHz fields has been reported in hippocampal network oscillations [39].

4.4 The nonlinearity is the detector: the σ'' curvature law and its controls

The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2 m$ across the operating range, vanishing and reversing sign at the inflection v_0 (Fig. 6)—demodulation is the sigmoid-curvature term of Eq. (10), and the closed-loop operating points sit where σ'' is large and negative.

In the recurrent column, the response scales quadratically at small drive (log–log slope 1.99, measured as in §3.1), and linearizing the field-receiving sigmoid reduces it from 1.34 to 0.0021 mV (Fig. 6b,c). The open-loop surrogate comparison in panel (d) supports its use for leading-order beat sweeps, with the waveform and near-Hopf limitations specified in §2.2.

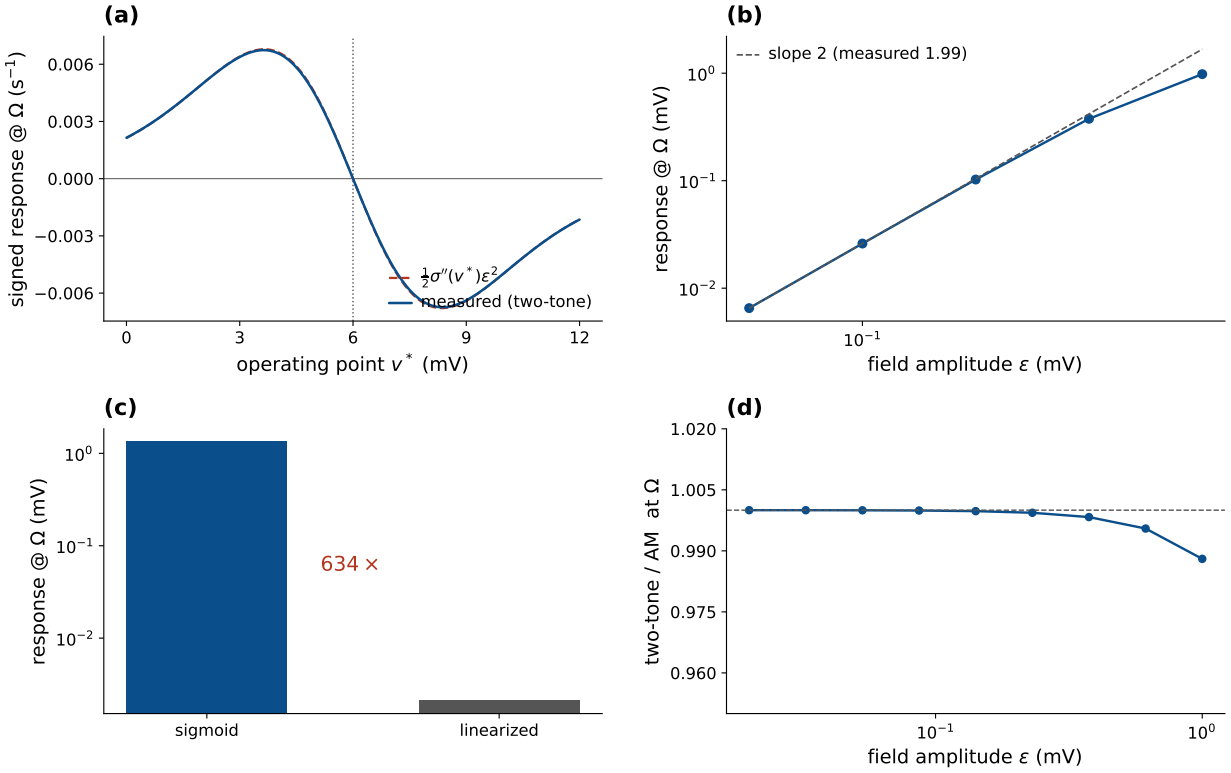


Figure 6: The curvature law, the square law, and the surrogate control (Jansen–Rit, genuine two tones). *(a)* The signed open-loop response tracks $\frac{1}{2}\sigma''(v^*)\varepsilon^2$ across the operating range (maximum deviation 0.9%), vanishing at the inflection v_0 and reversing sign across it. The mechanism is sign-agnostic: σ'' sets sign and phase while $|\sigma''|$ sets gain, so the resonance picture is unchanged up to an overall sign in the $\sigma'' > 0$ lobe, which this sweep spans. The sign flip is itself a prediction (§4.9). *(b)* The closed-loop response scales as ε^2, measured slope 1.99. *(c)* Replacing the field-receiving sigmoid by its tangent collapses the response 634×. *(d)* Surrogate control: the two-tone/AM ratio at Ω approaches 1 as $\varepsilon \to 0$ (1.0000 at 0.02 mV, 0.988 at 1 mV) for the open-loop detector at $v^* = 8$ mV. This checks the leading local source; it does not validate the recurrent AM-surrogate sweeps of Fig. 9 near the Hopf.

4.5 Exact mean-field confirmation: the NMM2 PING gamma generator

The exact PING mean field generates and amplifies the beat through its derived v_E^2 nonlinearity. With a 300 Hz two-tone field entering the excitatory current, the lock-in of r_E shows a growing gamma resonance below the Hopf and a response superposed on autonomous gamma above it (Figs. 8, S8). The resonance ridge intensifies through the Hopf; its appreciable frequency shift with operating point makes it diagonal, unlike the JR sweep shown here [31].

The receiver behavior thus also occurs with a nonlinearity derived from QIF spiking dynamics. Its coefficient differs from the sigmoid's: at 300 Hz the population lies on the fast side of its crossover, where carrier voltage excursions mix in the voltage equation. The fast-carrier approximation predicts the second-order response to within 1.1% at the four tested stable points (§2.3, Fig. 7).

4.6 What the mesoscale adds: coupling sets the gain at fixed detection

The resonance sweeps above drove the column toward the Hopf with the external input p, but p also shifts the operating point, entangling detection (σ'') with amplification. To isolate amplification, we vary the JR global connectivity C (coupling strength J) and solve analytically for the drive $p(C)$ that *pins* the field-free operating point v^*, so the detection gain $\sigma''(v^*)$ and the open-loop detector amplitude $A_{\text{open}} \simeq \frac{1}{2}|\sigma''(v^*)|\varepsilon^2 m$ are held constant. As $C \to C^*$ (the Hopf, $C^* \approx 137$) the damping $\gamma \to 0$, the demodulated response grows $\sim 30\times$ and the envelope resonance sharpens (Fig. 9a,b), a network effect since the local detector curvature is fixed while the recurrent transfer

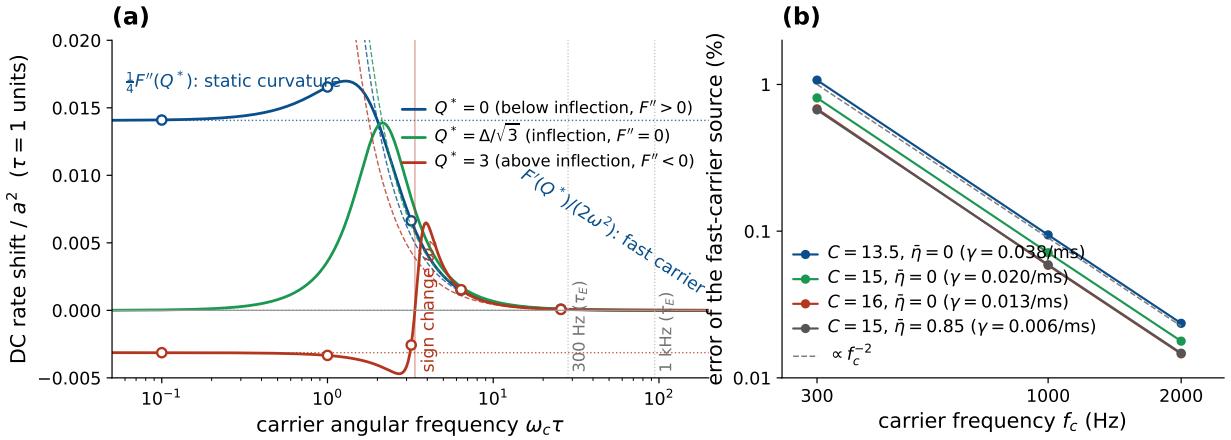


Figure 7: Static curvature at slow carriers, slope at fast ones. *(a)* Second-order DC rate shift of an uncoupled MPR population ($\Delta = 1$, $\tau = 1$) under a current carrier, Eq. (13), for operating points below, at and above the inflection of $F(Q)$. Dotted: the static limit $\frac{1}{4}F''$; dashed: the fast-carrier limit $F'/(2\omega^2)$; circles: RK4 integrations at $a = 0.05$ averaged over whole carrier periods (relative error below 10^{-3}). Above the inflection the sign changes at ω_z. The top axis converts $\omega\tau$ to a carrier frequency for $\tau = \tau_E = 15$ ms. *(b)* PING of Eq. (26): relative error of the fast-carrier difference-frequency source against the exact second-order response at four stable operating points (γ the damping of the gamma mode), two tones centered at 300, 1000 and 2000 Hz.

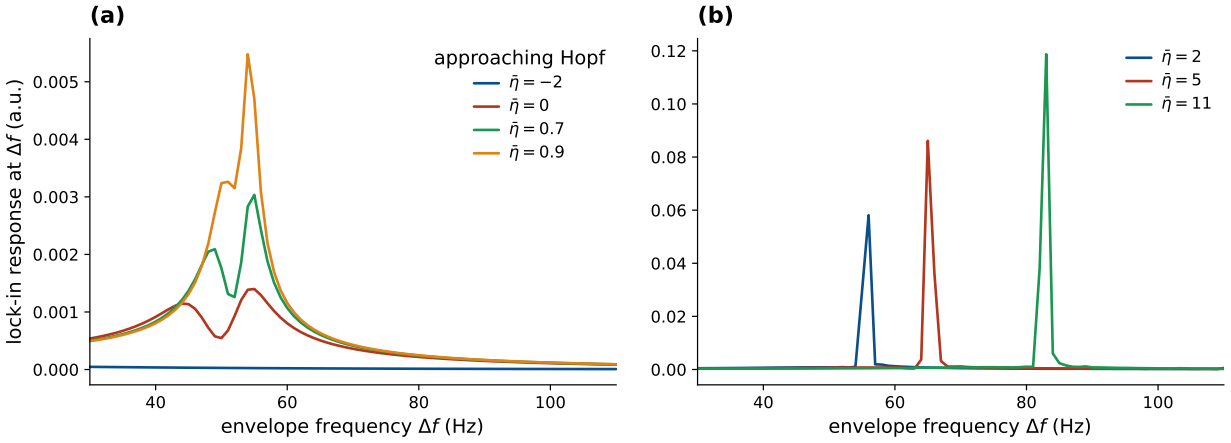


Figure 8: NMM2 PING gamma resonance (exact mean field, genuine two tones). Lock-in response of the excitatory rate r_E at the envelope frequency Δf under a two-tone TI field (carrier 300 Hz). *(a)* (stable focus, $\bar{\eta} < \bar{\eta}_H = 1.110$) forced resonance at the gamma frequency ($f_0 \approx 54$ Hz), the peak growing $0.0014 \to 0.0030 \to 0.0055$ as the Hopf is approached ($1/\gamma$ gain). *(b)* (limit cycle, $\bar{\eta} > \bar{\eta}_H$) response in the autonomous-gamma regime, the peak shifting upward with $\bar{\eta}$ ($56 \to 65 \to 83$ Hz). The input carries no power at Δf; the response is synthesized by the exact quadratic v^2 nonlinearity, with no sigmoid anywhere in the model. Matched AM and two-tone drives agree at Ω to 0.07% well below the Hopf and to 4% at $\bar{\eta} = 0.9$, consistent with the JR surrogate control (Fig. 6d).

changes. Its modal frequency and input–output projections vary with C as well as its damping. The rate-to-voltage gain A_Ω/A_{open} (mV s) tracks $1/\gamma$ before saturating near criticality (Fig. 9c); driving the same column to the Hopf through the input p rather than the coupling C gives a similar leading gain–γ trend. Both paths approach a weakly damped mode; their finite-distance gains need not coincide because the modal projections can differ.

The direct low-frequency control tests whether this amplification favors TI. For the AM surrogate with amplitude ε_{TI} and modulation depth m, and a tACS input of amplitude $\varepsilon_{\text{tACS}}$, the ratio of

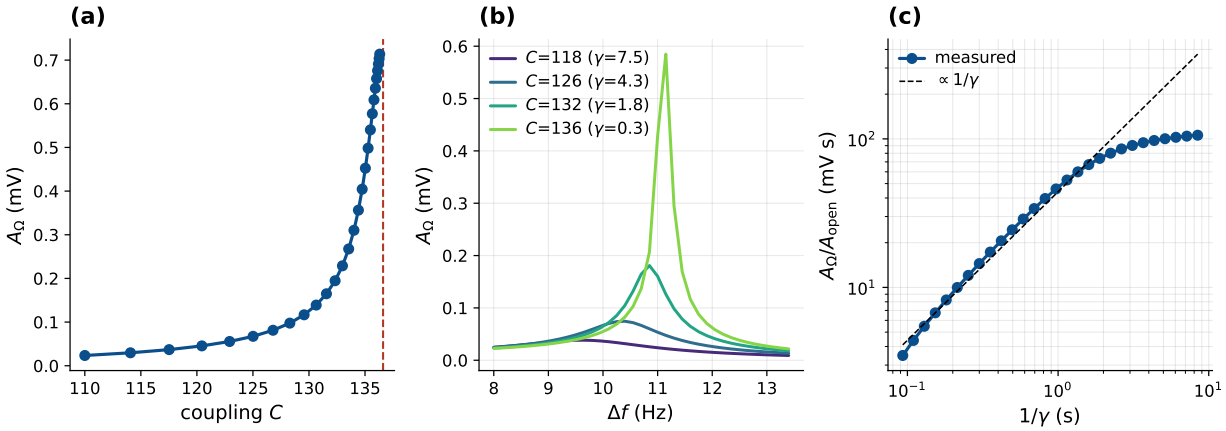


Figure 9: Coupling controls the amplification at fixed detection (Jansen–Rit). The input is the AM surrogate of Eq. (12). The field-free operating point v^* is pinned by co-solving $p(C)$, so $\sigma''(v^*) = -0.151$ and A_{open} are held constant throughout. *(a)* The demodulated A_Ω grows as the coupling $C\,(= J)$ approaches the Hopf C^* from the stable-focus side ($\gamma > 0$). *(b)* The envelope resonance sharpens and grows as $C \to C^*$. *(c)* The network gain A_Ω/A_{open}, in mV s, follows $1/\gamma$ (dashed) and saturates near the bifurcation.

voltage lock-in amplitudes in linear response is

$$\mathcal{R}_{\text{lin}} = \frac{|\sigma''(v^*)|\,\varepsilon_{\text{TI}}^2 m}{2\sigma'(v^*)\,\varepsilon_{\text{tACS}}}.$$

The common network transfer cancels, so the ratio has no dependence on γ when both responses are linear about the same state. At the amplitudes used in Figs. 9 and S3, $\mathcal{R}_{\text{lin}} = 0.143$. The cached surrogate sweeps give approximately 0.14 far from the Hopf, rising to 0.24 at $\gamma = 1.07\ \text{s}^{-1}$ and 0.47 at $0.12\ \text{s}^{-1}$. Genuine two tones give 0.14, 0.24, and 0.46, respectively, a change of at most 2.2% across the grid. These are response ratios for the stated unequal input amplitudes, not an equal-power comparison. In controls at three couplings, matching the tACS input to TI's leading beat source gives TI/tACS output ratios of 0.99, 0.94, and 0.92 for the surrogate, and 0.99, 0.93, and 0.90 for two tones. Increasing that direct input by a factor of 6.99 increases its response by 6.97, 3.95, and 1.96, respectively. These controls support earlier saturation of the larger response, accompanied by stimulation-induced mean shifts that field-free pinning does not cancel. The linear cancellation therefore does not extend to arbitrary finite-amplitude responses near criticality.

These sweeps approach the Hopf from field-free stable foci ($\gamma > 0$), so the $1/\gamma$ law concerns forced susceptibility. The frequency-dominance maps on the oscillatory side describe a different observable: their bands broaden with drive, but identifying an Arnold tongue requires phase tracking (§F).

In NMM2 PING, the coupling C is the mean inter-neuron QIF synaptic weight J. The leading fast-carrier source provides the normalization for this unpinned sweep (§2.3); finite-carrier source corrections over the full coupling range have not been bounded. Sweeping J at fixed background drive $\bar{\eta}$ toward the gamma Hopf J^*, the demodulated peak of r_E grows as $1/\gamma$ before departing from this scaling near criticality reveals two competing modes (Fig. S10), consistent with the . A second mode near 55 Hz carries the linear-response peak over $14.6 \lesssim J \lesssim 16.2$; the critical mode carries it at lower couplings and again near the Hopf, where its frequency is about 50 Hz. In that near-Hopf interval, $0.0014 \lesssim \gamma \lesssim 0.0081\ \text{ms}^{-1}$, the linear gain scales as $1/\gamma$ with a fitted exponent of 1.02 over a 5.8-fold damping range. The full-sweep exponent is 0.71, so a single-mode law does not describe the whole curve. This refines the coupling-controlled weak-field sensitivity of sensitivity expected in QIF networks [31]. Here : both coupling and rectifier are derived, and but the dominant mode can change.

At $A_f = 8$, the simulated response exceeds the normalized linear reference by up to 48% in the amplitude check, then saturates. At a quarter of that amplitude ($A_f = 2$, one sixteenth of the power),

the departures stay within 11% at the nine tested couplings. The normalization is stated in Methods; γ always describes the field-free equilibriumcritical mode, not the driven state. Although a DC-only averaged calculation predicts an unstable endpoint, the full periodically forced solution remains attracting: its largest perturbation multiplier is 0.034 over the 2 s forcing period. Increasing settling from 0.6 to 3 s changes the endpoint lock-in by 0.8%. The final points therefore describe a stable With genuine two tones, the 50.5 Hz endpoint control also attracts nearby trajectories: perturbations decay at -0.0014 ms^{-1}, and the 8 s Lyapunov estimate is -0.00079 ms^{-1}. These controls test a nonlinear driven responserather than , not linear gain about the DC-shifted equilibrium. The alpha JR and gamma NMM2 circuits share the leading gain scaling $1/\gamma$ gain scaling of their critical mode despite their different mixers, with amplification set by proximity to the Hopf and adjustable through coupling.

4.7 The resonance amplifies the envelope-locked modulation more than the mean rate

The demodulator feeds the network a constant term and a line at the beat Ω (Eq. (9)), and the transfer $G(\Omega)$ of Eq. (20) treats them differently: $G(\omega_0) \propto 1/(2\gamma\omega_0)$ diverges as the Hopf is approached, while $G(0) \propto 1/(\omega_0^2 + \gamma^2)$ stays finite. This comparison describes the response to the detector's DC and beat sources. The growing beat response itself rectifies and contributes another mean shift, quadratic in its amplitude. Before saturation that contribution can grow as $1/\gamma^2$, so bounded $G(0)$ does not imply a bounded mean shift throughout the approach to the Hopf. Pinning v^* to hold the local detector curvature fixed, as in Fig. 9, and driving the column toward the Hopf through the coupling, we read both quantities from the same observable, the pyramidal rate $r_P = \sigma(v + s)$ with the field in its argument. The With genuine two tones, its beat-locked modulation of r_P at $\Delta f = f_0$ grows 13.5× 13.2× (26 to 346339 mHz), tracking $1/\gamma$ before saturatingnear criticality; the shift of its mean grows too, 5.8× (−1.3 ; its mean shift grows 4.4× (−1.8 to −7.7 mHz), through rectification of the growing oscillation, but stays below 0.2% of the 3.9 Hz baseline (Fig. S11). Modulation exceeds mean shift by 19× The ratio of modulation to mean-shift magnitude rises from 14.5 far from the Hopf and by to 58 near $\gamma = 1$ s^{-1}, then falls to 44 closest to the Hopf as rectification of the amplified oscillation accelerates the mean shift. The surrogate, whose direct DC source is 3/4 of the two-tone value, gives ratios of 19, 66, and 45–66× near it. The network amplifies the envelope-locked modulation far more than the mean rate, so a . A small mean-rate change is the expected signature even where TI is acting, the mesoscopic counterpart of the can therefore accompany a much larger beat response in this weakly driven regime, consistent with the observation that TI alters spike timing more than firing rate in the primate brain [20]. The mean shift is small but nonzero, so the comparison must use commensurate measures, as here.

4.8 QIF responses distinguish beat modulation from field-induced gamma

The QIF network connects the population description to individual spikes [29, 26, 31]. With 8000 neurons per population and the same parameters as the PING mean field, it distinguishes a forced beat response from a stimulation-induced change of dynamical regime (Figs. 10, 11).

At $\bar{\eta} = 0.85$ and tone amplitude $A_f = 10$, the spiking network has a beat depth of 0.093 at 55 Hz and a mean-rate increase of 2.8%: modulation exceeds the fractional mean-rate change by a factor of 3.4. The mean field gives a larger depth, 0.128. Raising the spike cutoff from $V_{\text{peak}} = 100$ to 200 and 400 gives depths 0.127 and 0.139, while the mean-rate increase stays at 2.8–2.9%. At the original cutoff, doubling the population gives 0.083 and halving the time step gives 0.093. The response is therefore sensitive to the finite spike cutoff. Halving the step at cutoffs 200 and 400 gives depths 0.121 and 0.141, respectively: cutoff dependence persists, and quantitative convergence remains unestablished at this population size. At the plotted cutoff, the pooled spike-phase vector strength is 0.046; median per-neuron phase consistency remains negative in field-off and field-on windows at all three cutoffs. At $\bar{\eta} = 1.5$, the corresponding 42 Hz depth is smaller, 0.023, against a 1.6% mean-rate increase, and the existing gamma dominates the raster.

At the larger amplitude $A_f = 35$ and $\bar{\eta} = 0.85$, stimulation produces a dominant line near 52 Hz,

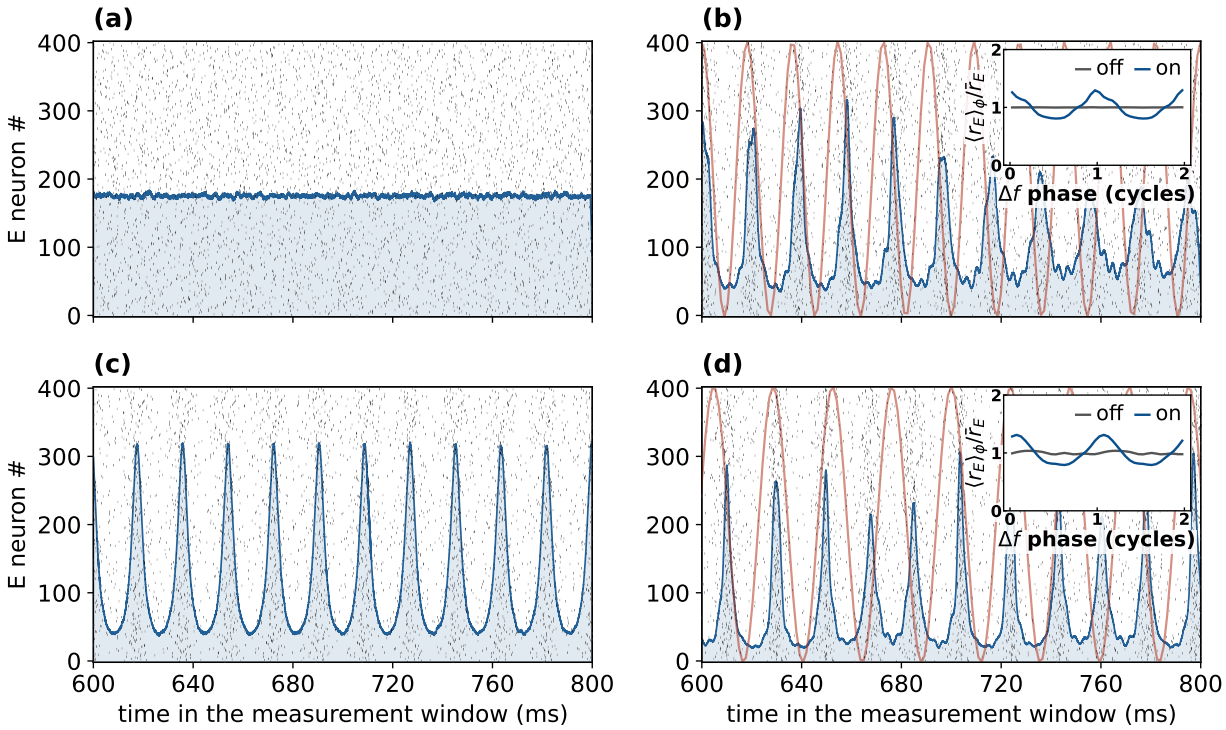


Figure 10: Beat-locked and gamma activity coexist in the QIF network. $N = 8000$ excitatory and inhibitory quadratic integrate-and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (26)). These are finite-amplitude examples ($A_f = 35$ per tone, carrier $f_c = 300$ Hz). Black ticks are E spikes (400 of 8000), blue is the smoothed population rate on a separate scale, and red is the TI envelope in arbitrary units. *(a,b)* Field-free stable state ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz. The field-on response is dominated by field-induced gamma near 52 Hz (depth 0.70), with a smaller beat component (depth 0.29). *(c,d)* Field-free oscillatory state ($\bar{\eta} = 1.5$), TI off then on at $\Delta f = 42$ Hz. Gamma near 53–54 Hz dominates the raster, with depth about four times the beat component's 0.23; their coexistence gives slow variations in burst height. Depth is line amplitude divided by mean rate. The final 200 ms of the common measurement interval are shown. *Insets* in *(b,d)* fold the rate on beat phase over 44 and 34 cycles (off gray, on blue), normalized to each condition's mean. This averaging suppresses non-beat-locked gamma and displays the smaller beat component on a common 0–2 scale. The fold alone does not establish gamma-amplitude gating or phase capture.

with modulation depth about 0.70, and a smaller component at the 55 Hz beat, with depth 0.29. The second-order carrier-averaged system predicts an excitatory drive shift $\delta\eta_E \approx 1.65$, beyond the E-only Hopf threshold $\delta\eta_E^* \approx 0.286$, with a leading eigenvalue whose real part is $+0.032$ ms^{-1}. Direct mean-field integration reproduces the off-beat line, supporting the interpretation of panel (b) as field-induced gamma with a smaller forced component.

The equal-RMS unmodulated carrier provides a control for this distinction in the mean field (Fig. 11c,d). It produces a rectified E-drive shift within 3% of TI's and a similar mean-rate increase. At $A_f = 35$, both waveforms induce gamma near 51 Hz, whereas TI also produces the 55 Hz beat component. Increasing the measurement window from 0.8 to 3.2 s reduces the carrier's apparent 55 Hz depth from 0.011 to 9.2×10^{-5}, consistent with spectral leakage, while TI's beat depth remains 0.30–0.31. At $A_f = 10$, neither induces gamma, and TI alone produces a resolved beat response. The control separates induction of an oscillation from its beat-frequency response; it has not been run in the spiking network.

At $\bar{\eta} = 1.5$, gamma activity is already present without stimulation. With the field on, the rhythm near 53–54 Hz remains about four times larger than the 42 Hz beat-locked component. Gamma therefore dominates the raster in panel (d); the coexistence of the rhythms produces slower variations in burst height. Folding the rate on the beat phase suppresses the non-beat-locked gamma and exposes the smaller component. The inset shows that phase average, not the full rate variation.

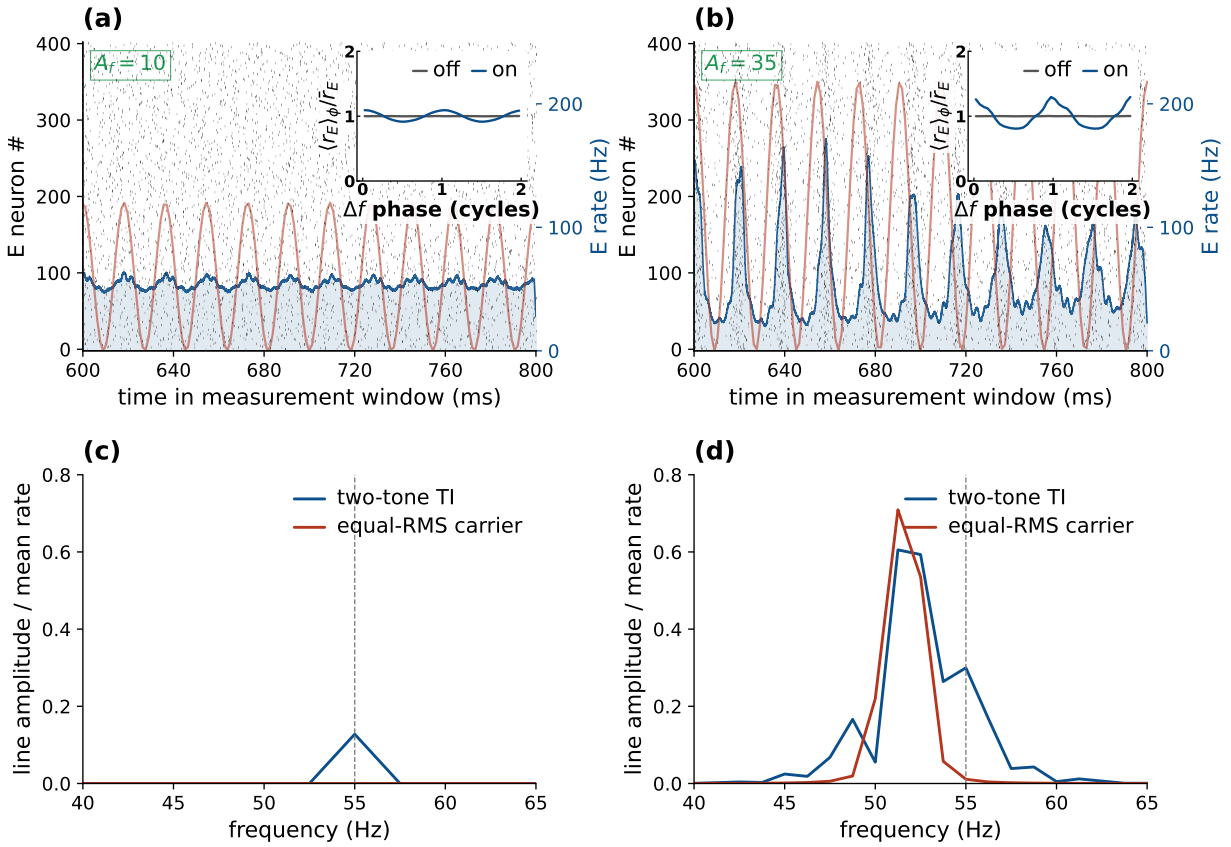


Figure 11: Beat modulation below and across the induced-gamma transition. All panels use $\bar{\eta} = 0.85$, carrier 300 Hz, and TI beat 55 Hz. *(a,b)* Spiking-network responses at $A_f = 10$ and 35 per tone, respectively: black ticks are spikes from 400 of 8000 E neurons; blue is the smoothed population rate on the common right-axis scale, and red is the envelope in arbitrary units. Insets show phase-folded rate divided by its mean (off gray, on blue), on the same scale. The beat depths are 0.093 and 0.290, and the mean-rate increases are 2.8% and 10.5%. At the larger amplitude, non-beat-locked gamma dominates the raster and is suppressed by the fold. *(c,d)* Corresponding mean-field spectra under TI (blue) and an equal-RMS single carrier (red), whose peak amplitude is $\sqrt{2}A_f$. Line amplitude is normalized by each trace's mean; the dashed line marks 55 Hz. Spectra use a Hann window over 800 ms after 800 ms settling. Both waveforms induce gamma at the larger amplitude, while TI produces the additional beat response. The carrier's apparent 55 Hz line falls below 10^{-4} with a 3.2 s window, while TI's persists. The carrier comparison is a mean-field result. The spiking panels retain $V_{\text{peak}} = 100$; at cutoffs 200–400, the $A_f = 10$ depth ranges from 0.121 to 0.141 across the two tested steps (§4.8).

The population spectrum also contains a sum sideband near 95 Hz and a smaller difference component near 11 Hz (Fig. S12b). These lines establish nonlinear mixing, without uniquely identifying gamma-amplitude gating or phase modulation. The field-free gamma spectra of the spiking network and exact mean field agree in frequency (Fig. S12a).

At $A_f = 35$, the mean rate rises by about 8–11% across the two conditions, while the beat modulation depth, $A_\Omega / \langle r_E \rangle$, rises from below 0.003 to 0.29 ($\bar{\eta} = 0.85$) and 0.23 ($\bar{\eta} = 1.5$) (Fig. S12c), measured over 44 and 34 whole beat cycles after 800 ms of settling (Methods). Timing modulation therefore exceeds the relative mean-rate change by a factor of about three at the population level. Per neuron the locking is weak: the pairwise phase consistency of single-neuron spike phases at Δf rises from its field-free control (median −0.02) to a median of 0.003–0.005, and the pooled vector strength from below 0.02 to 0.12–0.13. The population's beat-locked component is built from many weakly biased cells. This gives a qualitative counterpart of the timing-versus-rate distinction in §4.7 and in primate recordings [20]; the present amplitudes do not establish the size of that effect under weak physiological stimulation.

Beat-locked and gamma activity coexist in the QIF network. $N = 8000$ excitatory and inhibitory quadratic integrate-

and-fire neurons whose exact mean field is the NMM2 PING circuit (Eq. (26)). These are finite-amplitude examples ($A_f = 35$ per tone, carrier $f_c = 300$ Hz). Black ticks are E spikes (400 of 8000), blue is the smoothed population rate on a separate scale, and red is the TI envelope in arbitrary units. *(a,b)* Field-free stable state ($\bar{\eta} = 0.85$), TI off then on at $\Delta f = 55$ Hz. The field-on response is dominated by field-induced gamma near 52 Hz (depth 0.70), with a smaller beat component (depth 0.29). *(c,d)* Field-free oscillatory state ($\bar{\eta} = 1.5$), TI off then on at $\Delta f = 42$ Hz. Gamma near 53–54 Hz dominates the raster, with depth about four times the beat component's 0.23; their coexistence gives slow variations in burst height. Depth is line amplitude divided by mean rate. The final 200 ms of the common measurement interval are shown. *Insets* in *(b,d)* fold the rate on beat phase over 44 and 34 cycles (off gray, on blue), normalized to each condition's mean. This averaging suppresses non-beat-locked gamma and displays the smaller beat component on a common 0–2 scale. The fold alone does not establish gamma-amplitude gating or phase capture.

4.9 Falsifiable predictions and tests

The mechanism makes four predictions, each with an experimental handle.

1. **Carrier dependence separates access from mixing.** In the static limit and in the fast-carrier QIF asymptote, the leading beat is independent of f_c at fixed local polarization, provided the carrier does not excite another network mode. At fixed applied field, access restores frequency dependence (Eq. (22)); detector dynamics can add it near the crossover. *Test:* measure or estimate local polarization during a carrier sweep and compare the response before and after correcting for access. The remaining frequency dependence constrains the mixing mechanism. Its cutoff is to be measured, not assigned to a particular compartment in advance.

2. **State- and carrier-dependent response.** The generated beat depends on the operating point through the detector coefficient, and the two modeled detectors disagree on how: the memoryless sigmoid follows $\sigma''(v^*)$ and reverses sign across the inflection, where PEIX (Eq. (24)) peaksis zero; the dynamic mean field at a fast carrier mixes in the voltage dynamics and is read through the rate and network susceptibility; its DC rectification follows F', without the same curvature sign rule (§2.3). *Test:* vary tonic drive and carrier frequency while measuring both amplitude and phase of the beat response. Compare these with the predicted mixing coefficient after accounting for changes in network susceptibility; a response-amplitude change alone does not isolate the detector.

3. **Transduction class.** A smooth memoryless response with nonzero curvature transduces A^2 at weak input, whereas an ideal envelope follower transduces A (§2.4), and the two are separable at a single matched target. The former gives $Y_\Omega \propto \varepsilon^2$ with a second-harmonic ratio vanishing as $\varepsilon \to 0$; the latter gives $Y_\Omega \propto \varepsilon$ with the ratio pinned at $1/5$. A threshold detector instead has an onset at $2\varepsilon = \delta$ and an amplitude-dependent harmonic ratio. *Test:* sweep field amplitude off resonance (so that both lines sit on the flat part of G) and read the log–log slope together with $Y_{2\Omega}/Y_\Omega$; on resonance the network suppresses the second harmonic by $\sim 3\mathcal{Q}$ and the test must be corrected by an independently measured $G(2\Omega)/G(\Omega)$ (§3.1).

4. **Frequency selectivity and state-dependence.** The response is largest when Δf matches a region's resonant rhythm and is amplified as $1/\gamma$ near a bifurcation. Changes in operating state can therefore alter both the generated drive and the recurrent gain [24]. *Test:* sweep Δf through a region's intrinsic band across brain states and track the resonant peak and its width.

Table 3 compares the predictions and related model consequences with existing evidence. Two are qualitatively consistent with published observations: TI alters spike timing more than firing rate in vivo [20] (our $G(\omega_0) \gg G(0)$ asymmetry, the preferential beat response and its finite-amplitude limits in §4.7–4.8), and most pyramidal TI responses vanish when synaptic transmission is blocked [16] (a network read-out, which tests network involvement rather than excluding a surviving cellular detector). A quantitative reanalysis of these datasets against the model is a natural next step, and §5 specifies the amplitude-exponent fit it would require.

Prediction (model basis)	Existing evidence	Proposed test
Transduction class: $Y_\Omega \propto \varepsilon^2$ and vanishing $Y_{2\Omega}/Y_\Omega$ (square law) vs. $\propto \varepsilon$ and 1/5 (envelope following); threshold detection has an onset and variable ratio	— (untested; existing comparisons match extracellular field, not ε)	Amplitude sweep off resonance; transfer-corrected harmonic ratio
Timing-dominated response: the envelope-locked modulation of the rate grows toward the Hopf faster than its mean ($G(\omega_0) \gg G(0)$) beat modulation can exceed the mean shift; their ratio peaks and then falls as the amplified oscillation rectifies	*Qualitatively consistent*: TI alters spike timing more than firing rate, primate [20]	Lock-in of spike timing at Δf vs. the mean-rate change, in the same effect measure
Network read-out (population nonlinearity + synaptic recurrence)	*Qualitatively consistent*: most pyramidal TI responses vanish under synaptic block [16]	Synaptic block should reduce recurrent amplification; cellular mixing may persist
Carrier independence at fixed polarization (no ω_c in A_Ω)	TI thresholds inherit the carrier's f_c-dependence [21, 22]; bare kHz weakly effective [39]	Fix in-zone polarization, sweep $f_c \to$ flat envelope response in the stated limits; fit access and residual detector dependence
State- and carrier-dependent response (the detector coefficient is model-dependent)	Cell-type-specific TI responses (PV vs. pyramidal) [16]	Shift v^* by state at fixed beat; sweep f_c at fixed beat
Frequency selectivity, sharpened $\sim 1/\gamma$ near criticality	Quantitative gain–bandwidth relation not tested here in biological data	Sweep Δf through a region's band across brain states

Table 3: The mechanism's predictions confronted with existing in-vivo / in-vitro evidence. Entries cite documented qualitative findings; quantitative reanalysis is left to future work.

5 Discussion

Axonal ion-channel rectification is already demonstrated in single-neuron TI models [2], with axonal excitation and block also found in cortical-neuron models [9]. Our population representation connects candidate fast access to cellular mixingand recurrent amplification through a short effective membrane time. Cohen's periaxonal circuit motivates separating axonal from somatic voltage dynamics [7]; nodes and the sodium-channel-rich AIS are candidate access sites [8]. Luff et al.'s subthreshold mixing establishes a cellular detector [10], without localizing it to either candidateThe calculations separate the field-to-membrane transfer, the nonlinear source it generates, and the circuit response to that source. Esmaeilpour et al. [6] already related network TI sensitivity to short, axon-like membrane times and distinguished sensitivity from selectivity mediated by network adaptation. Here the exact QIF mean field connects static and dynamic mixing, the weak-input product law determines spatial weighting, and matching the generated drive exposes the linear amplifier shared by TI and tACS. Fast access, nodal rectification, and the AIS candidate retain the distinct anatomical and cellular evidence described in the Introduction.

At applied extracellular fields of order 1 V/m, voltage-gated channels respond through membrane polarization, not through direct action of the field on the protein: the structural effects reviewed by Cifra et al. [40] are reported at fields from 10 kV/m to GV/m, with thresholds that depend on the protocol and molecular system. That review places fields below 0.1 kV/m outside its scope, where it expects direct effects to be negligible against thermal noise. Presynaptic terminals can

polarize more strongly than somata [41, 42, 43], but small size alone does not imply fast charging: for an isopotential membrane, the product of specific resistance and capacitance is independent of area. Conductance, geometry, and the mode of stimulation determine the response.

The measurements in Table 2 and related models support testing shorter access times, with their TI transfer still to be calibrated. The 1 and 0.4 ms network sensitivity cases of [6] precede the present effective-time hypothesis. Luff et al.'s pulse-width-modulated TI (PWM-TI) model uses a 1 ms passive response within a 1–100 ms parameter sweep [44]. Separately, Semenov et al. [23] measured field-induced membrane charging near 0.7 μs in isolated neurons under strong, microsecond field pulses. This is a different response mode from intracellular current charging or periaxonal conduction. Together these studies motivate examining a shorter effective access time, while leaving its value and field coupling preparation-dependent. A 10–100-fold reduction in a QIF membrane time is therefore a sensitivity hypothesis. Its population-level consequence can be studied through the existing slow-synapse limit, with each physical filter included once; a new periaxonal circuit is not needed for that limiting argument.

Once the field reaches a smooth nonlinearity with a nonzero second-order term, weak-input cellular contributions combine into a quadratic mixing coefficient. This leading-order account does not require identifying each channel; QIF provides a derived realization. It connects cellular mixing [2, 10] with the network dependence seen in slices and simulations [16, 17, 18][16, 17]. Cellular mixing and circuit amplification are therefore not competing explanations.

Weak amplitude alone does not make that coefficient static. Fast population relaxation relative to synapses gives the NMM2 slow-synapse limit; relaxation must also be fast relative to the carrier for curvature to describe its mixing. Outside that limit the coefficient is dynamic. Its magnitude, phase, and frequency dependence remain model-specific, while the static slope converts slow input into rate and enters recurrent gain (§2.1).

A spatial TI dose map depends on this physiology. The usual envelope-modulation map assumes a response determined by envelope excursion; the square-law map instead weights it by the larger local polarization (Eq. (18)). Its accompanying DC source follows the summed carrier intensities under the conditions of §2.4, providing an operating-point shift that the equal-RMS carrier control isolates and that can cross the population Hopf (§4.8). Orientation, access, operating point, and network susceptibility further shape the response. A sharp rectifier can approximate envelope following over a finite amplitude range; smoothing or heterogeneity can restore quadratic behavior at smaller inputs. The amplitude and harmonic tests in §2.4 3.1 distinguish these regimes.

Stimulation intensity can also change the mechanism. At high intensities, TI and unmodulated kHz fields can have similar excitation thresholds, with modulation patterning firing rather than lowering the threshold [21, 23]; some peripheral responses do not require envelope extraction [22]. The weak-input calculation here concerns modulation of ongoing activity through mixing and amplification [20, 45]. Direct carrier-evoked spiking and conduction block are outside its scope. The finite-amplitude QIF examples illustrate a related distinction: rectification can shift a stable population across its Hopf and induce gamma activity alongside the beat-locked component. The driven component must be separated from that change in regime. The boundary depends on the initial state's distance from the bifurcation as well as on field amplitude. These examples and the laminar extension describe behavior beyond the weak-input expansion; they do not establish physiological thresholds.

The recurrent amplifier is shared with direct low-frequency stimulation. In our coupling sweep, holding the JR field-free operating point fixed holds the local curvature fixed while the network gain changes. A direct in-band drive shows the same $1/\gamma$ growth and resonance sharpening (Fig. S3), consistent with the known resonant susceptibility of recurrent excitatory–inhibitory networks [15, 31]. Near the Hopf, this is temporal critical slowing in a mean-field model, not evidence of a diverging spatial correlation length. Related models already reproduce tACS entrainment and

AM-driven phase locking [46, 47, 48]. The separation here is between the drive generated by mixing and the response to that drive. When TI and tACS are matched for effective in-band input, observable, and operating state, they share the linear amplifier. Within linear response, resonance alone therefore gives TI no efficiency advantage. At finite amplitude the response ratio can change near the Hopf, as the coupling sweep demonstrates (§4.6); its value then depends on drive amplitude and the driven state as well as on field-free damping.

Absolute response estimates remain sensitive to access and transduction. Measured static DC field-to-voltage couplings are 0.12–0.210.18 mV per V/m in hippocampus (0.21 at 10 Hz AC) [49, 50] and reach about 0.5 mV per V/m in cortical L5/6 pyramidal cells [51]; human cortical fields near 0.8 V/m have been reported at 2 mA [52, 53]. These static couplings do not fix the coupling at kHz. For equal TI carrier polarizations ε, the JR beat drive is $\sigma''\varepsilon^2/2$, whereas a direct drive of the same peak polarization gives $2\sigma'\varepsilon$. Their leading amplitude ratio is $\eta_{\mathrm{det}} \simeq |\sigma''|\varepsilon/(4\sigma')$: the relative efficiency decreases as the input is weakened in this perturbative regime. A larger recurrent gain amplifies both drives.

For scale, a reverberation time $\tau_{\mathrm{rev}} = 1/\gamma$ gives a near-Hopf quality factor $\mathcal{Q} \simeq \pi f_0 \tau_{\mathrm{rev}}$ $\mathcal{Q} = \pi f_0 \tau_{\mathrm{rev}}$ and power bandwidth approximately $1/(\pi\tau_{\mathrm{rev}})$. Population relaxation over a few hundred milliseconds [54] would give a quality factor of order ten at 10 Hz and a bandwidth of order 1 Hz. This is an order-of-magnitude comparison: the measured relaxation is not specific to the alpha mode. Narrow amplification also makes detuning consequential.

Primate measurements illustrate both the relevance and the the limits of an efficiency comparison. Vieira et al. [20] used modulation depths of 0.5–0.62 V/m and reported TI effects at reported two distinct experiments. Their two-pair TI comparison gave a phase-locking effect ratio of 0.34 times those of conventional against tACS (95% confidence interval 0.04–0.65). Their within-electrode AM comparison , 65 of 154 neurons), without correcting for TI currents up to ± 2.5 mA per pair versus ± 1 mA for tACS. A separate single-pair AM-tACS control compensated for frequency-dependent current shunting and matched shunting to match extracellular modulation depthnear the recorded cells. A favorable fast-access estimate with $\varepsilon \approx 0.1$ mV gives only about 0.8% for the JR detector, below the reported ratio. This discrepancy is not a calibrated rejection of that detector: the comparison involved twenty neurons, with ten responding to at least one . Of twenty cells, ten responded to either modality and seven to both; in the latter group, median phase-locking value (PLV) depends on baseline locking and network state; and the matched extracellular fields do was 0.10 for AM-tACS and 0.20 for tACS, against 0.043 at baseline. The reported TI modulation depths were about 0.5–0.62 V/m. These measurements show a weaker AM effect, but matched extracellular depth does not ensure matched polarization at the cellular mixer. An unspecified nonlinear relation between generated drive and PLV also prevents identification of the input-amplitude exponent. The data establish a weaker AM effect than , and PLV depends on baseline locking and state.

For an illustrative fast-access budget, take a per-tone field $E_j = 0.5$ V/m, $\lambda_0 = 0.5$ mV per V/m, $f_c = 2$ kHz, and the assumed $\tau_m = 0.2$ ms. Then $\varepsilon = \lambda_0 E_j |H_m| \approx 0.092$ mV, rounded to 0.1 mV for scale; this is an assumed coupling budget, not an inferred polarization in the recorded cells. At $v^* = 8.30$ mV the JR detector gives $\eta_{\mathrm{det}} \approx 0.8\%$ for 0.1 mV, below the effect ratio in the matched-depth low-frequency drive , but do not isolate the transduction lawAM-tACS control. Generated drive and PLV are different observables, so their comparison neither calibrates this detector nor identifies the input-amplitude exponent.

For comparison, the matched-peak efficiency over the logistic family has a supremum of approximately 0.348 (App. D). Approaching it at weak fields requires a transfer curve narrowed to microvolt scales; the limiting detector is step-like and has a second-harmonic ratio near 0.15, distinct from both weak square-law detection and linear envelope following. This bound applies to that memoryless family and matching convention, not to the dynamic QIF detector or to all cellular mechanisms. Its proximity to an experimental ratio is not a fitThe unequal-current TI ratio and the matched-depth AM-tACS control do not measure this matched-peak, open-loop ratio.

Quantitative physiological predictions require calibration of access and mixing. Their values are not fixed by the quadratic scaling law. In comparisons of stimulus-evoked firing, Hodgkin–Huxley and Frankenhaeuser–Huxley models reproduce responses missed by the tested leaky, exponential, and adaptive-exponential integrate-and-fire models [55]. Such model dependence concerns the specific response, without invalidating the weak-input expansion. Noise, heterogeneity beyond the assumed excitability distribution, and interactions between columns can broaden or shift the deterministic resonances. Cascading ordinary synaptic filters cannot solve carrier access, because they attenuate the carrier before a subsequent nonlinearity receives it. Human results are also mixed, including reported peripheral effects without central effects [56] and null occipito-parietal alpha-power modulation [57]. Testing the proposed mechanism requires measuring access, generated beat, and network tuning in the same preparation.

6 Conclusion

TI requires field-to-membrane access, nonlinear detection, and amplification. Building on existing axonal TI mechanisms, we represent candidate fast access through nodes and the AIS by a short effective membrane time, with periaxonal circuitry motivating separate axonal and somatic responses. Their TI transfer and the response at human field strengths require calibration.

For weak perturbations of ongoing activity, smooth nonlinearities with a nonzero second-order response generate the beat through effective quadratic mixing. The population models realize this description and amplify the generated drive near a network resonance. A large response alone establishes neither efficient detection nor the microscopic mixer.

Fast population relaxation gives the NMM2 slow-synapse limit, a nonlinear static rate map with dynamic synapses. Its curvature describes carrier mixing only when relaxation is also fast relative to the carrier; otherwise the dynamic response applies. Fast access and slow recurrent resonance can coexist, with each physical filter represented once.

In the weak square-law regime, beat generation depends on the product of the effective carrier inputs, not only on the envelope excursion. TI and a direct low-frequency drive share the downstream amplifier when input, observable, and state are matched and both responses remain linear, so resonance alone gives TI no efficiency advantage. Finite-amplitude responses near the Hopf can depart from that ratio. Measuring local carrier polarization, the generated beat, and circuit gain separately would calibrate the stages of the proposed mechanism.

Code and data availability

Simulation code, cached results, and figure-generation scripts are provided at the paper's GitHub repository; the public archive is linked through Zenodo (doi:10.5281/zenodo.21009618). The repository `README` maps figures to their scripts and distinguishes recomputation from replotting cached data. A reconstructed producer with explicit run settings reproduces every archived spectral bin of the two-tone kHz illustration (Fig. S6) exactly. The LaNMM setup is a drawn schematic. The LaNMM carrier comparison has both measurement-window caches and a reproducible summary script. The code is released under CC-BY-4.0.

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Competing interests

Giulio Ruffini, Borja Mercadal, Alex Just, and Francesca Castaldo are employees of Neuroelectrics, which develops and sells transcranial electrical stimulation devices. Giulio Ruffini is a co-founder and shareholder of Neuroelectrics. The other authors declare no competing interests.

Author contributions

Using the CRediT taxonomy: **G. Ruffini**: conceptualization, methodology, formal analysis, software, investigation, visualization, writing (original draft), supervision, funding acquisition. **B. Mercadal**: validation, visualization, writing (review and editing). **A. Just**: software, visualization, writing (review and editing). **R. de Palma Aristides**: software, visualization, writing (review and editing). **F. Castaldo**: conceptualization, writing (review and editing). **N. Grossman**: conceptualization. **S. Canals**: supervision, writing (review and editing). **C. Mirasso**: supervision, writing (review and editing).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Anthropic Claude and OpenAI ChatGPT to streamline the narrative and to assist with code and figure generation from data the authors produced. The authors reviewed and edited all resulting content and take full responsibility for the content of the published article.

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